

浙江大学

本科生毕业论文(设计)

文献综述和开题报告



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年级与专业 2022 级 电子科学与技术

所在学院 信息与工程学院

一、题目：面向手势识别的 Doppler 雷达传感技术

二、指导老师对文献综述、开题报告、外文翻译的具体要求

1. 文献综述和开题报告要求：

- (1) 查阅国内外相关文献资料 10 篇以上（其中外文文献不少于 5 篇），近 5 年文献不少于 3 篇。在此基础上完成文献综述，正文要求 4000 字以上。
- (2) 完成外文翻译，正文（中文）要求 3500 字以上。
- (3) 完成开题报告，正文要求 4500 字以上。

2. 文献综述要求：

- (1) 掌握多普勒信号的特征与处理方法，了解其在生物探测领域的应用价值、技术优势与局限，以及国内外研究进展。
- (2) 综述现有手势信号特征提取与分类算法，分析其国内外研究现状。
- (3) 对比各类手势识别技术，阐明采用多普勒雷达实现手势识别的研究意义。
- (4) 做好毕业论文相关知识的准备工作。

3. 开题报告要求：

- (1) 做好毕业论文相关知识的准备工作。
- (2) 给出基于多普勒雷达的手势识别系统设计方案，选定核心处理算法。
- (3) 对论文要完成的工作有清晰的认识，给出具体的研究方案，预见可能遇到的问题和困难。
- (4) 提出论文的研究成果或预期达到的目标。
- (5) 安排好论文的进度，对于每个阶段的工作有详细的规划。
- (6) 给出本人已进行的研究或设计工作及已取得的结果。

指导教师（签名）



2025 年 11 月 25 日

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一、文献综述

1 背景介绍

1.1 概要

在普适计算与物联网 (IoT) 飞速发展的时代背景下, 人机交互 (HCI) 模式正经历着从传统的物理接触式向自然非接触式的深刻变革。传统的交互终端, 如多点触控屏幕、物理按键及鼠标, 虽然在精度和响应速度上已趋于成熟, 但在诸多特定应用场景中表现出局限性^[1]。例如, 在医疗手术室、无尘车间或驾驶舱等环境中, 物理接触不仅可能带来卫生隐患, 还可能分散操作者的注意力, 降低工作效率。随着增强现实 (AR)、虚拟现实 (VR) 以及智能座舱等技术的不断演进, 用户对于更直观、更自由、更符合人类表达直觉的交互方式提出了更高要求。手势识别作为人类非语言交流的核心组成部分, 因其具备跨越语言障碍、支持隐蔽操作及高自由度表达等特性, 已成为下一代人机交互接口的研究焦点^[2]。

手势识别技术路线主要分为视觉传感、穿戴式传感与射频传感三类。以摄像头为核心的视觉方案虽然在图像识别领域取得了显著突破, 但其高度依赖光照强度, 且在暗光环境、视距 (LoS) 遮挡及涉及用户私密空间的场景中, 面临严重的性能衰减与隐私泄露风险。穿戴式方案如数据手套、肌肉电信号 (sEMG) 传感器及惯性测量单元 (IMU), 虽能提供厘米级的运动追踪精度, 但其物理束缚感、繁琐的佩戴流程以及高昂的设备维护成本, 限制了其在普通消费电子领域的普及^[3]。相比之下, 基于微波与毫米波 (mmWave) 的多普勒雷达传感技术, 凭借其全天候工作、穿透遮挡能力及天然的隐私保护优势, 展现出极高的学术价值与应用潜力^[4]。

雷达传感器通过主动发射电磁波并接收目标反射的回波, 能够捕捉目标在微米至厘米级别的微波波动。多普勒效应作为其核心物理机制, 将手部及手指的复杂运动转化为回波信号中的频率偏移与相位调制。近年来, 随着半导体制造工艺向 28nm 及更深纳米级演进, 毫米波雷达芯片正朝着超大规模集成 (SoC)、低功耗及微型化方向跨越式发展。Google 的 Soli 项目首次展示了将 60GHz 调频连续波 (FMCW) 雷达集成于智能手机的可行性, 标志着雷达手势识别从军事、气象等领域正式跨入个人终端消费市场^[5]。我们综合讨论近些年来的主流手势识别技术的核心性能指标如表1.1。

与此同时, 随着人工智能技术 (Artificial Intelligence, AI) 的迅猛发展, 更

表 1.1 各类传感技术评估维度对比

评估维度	多普勒雷达 (mmWave)	计算机视觉 (RGB-D)	惯性传感器 (Wearable)	超声波/声学 感知	Wi-Fi CSI 感 知
隐私安全 性	极高：无图 像、无生物 识别信息	低：存在面 部及隐私环 境泄露风险	中：仅运动 数据，但需 数据存储	高：声波无 法还原图像	高：射频信 号难以直观 解读
环境适应 性	极强：不受 光照、烟雾、 遮挡影响	弱：光照敏 感，视距受 限	强：不受外 界环境干扰	中：受环境 噪声和气流 影响	强：支持穿 墙探测
佩戴便捷 性	免佩戴：真 正的非接触 式交互	免佩戴	差：需要物 理接触，舒 适度低	免佩戴	免佩戴
分辨率/精 度	高：可捕捉 亚毫米级微 动	极高：像素 级几何形状 识别	极高：高精 度采样	中：依赖声 速，动态响 应慢	低：通常为 厘米级粗糙 识别
硬件成本	中等：由于 CMOS 集成 度提高持续 下降	低：手机摄 像头普及	中-高：需额 外硬件	极低：利用 现有麦克 风/扬声器	极低：复用 路由器硬件

多智能终端在传统的用户交互的基础上，集成了多种传感器和智能算法，实现了更为复杂和智能化的交互方式。在当下的分类和预测任务中神经网络（Neural Network）能够处理图像、语音和文本等高度非线性的原始数据，并将其映射到预定义的类别中，例如：多层感知机（MLP）作为基础架构，通过全连接层对结构化数据（如用户画像、财务指标）进行特征组合，实现精准的二分类或多分类^[6]；卷积神经网络（CNN）在图像识别和目标检测领域表现卓越，它通过卷积核提取空间特征，能够自动识别复杂的视觉模式，用于人脸识别、医疗影像诊断等领域^{[7][8]}。循环神经网络（RNN）及其变体（LSTM/GRU）专注于序列分类，能够捕捉文本或语音中的上下文语义^[9]。

1.2 问题建模

该系统由两个核心部分组成：雷达数据采集部分与神经网络识别部分。整体流程通过嵌入式处理器/MCU 进行统筹管理。示意图见1.1。数据采集部分使用毫米波天线阵列（mmWave Antenna Array）向空间发射电磁波信号。当手部进行动态动作时，电磁波发生反射，由接收天线（Rx）捕捉回波信号。回波信号首先经过模拟前端。主要包括利用低噪声放大器（LNA）对微弱回波进行放大，并通过混频器（Mixers）进行下变频处理。处理后的模拟信号通过模数转换器（ADC）转化为数字信号，进入数字处理阶段。随后在神经网络处理阶段原始数字信号被转化为神经网络可理解的高维特征，提取出的特征（如 RDM 和微多普勒谱图）被送入分类器模型，系统输出识别结果（如“滑动”、“推”等动作标识），并触发相应的指令响应。

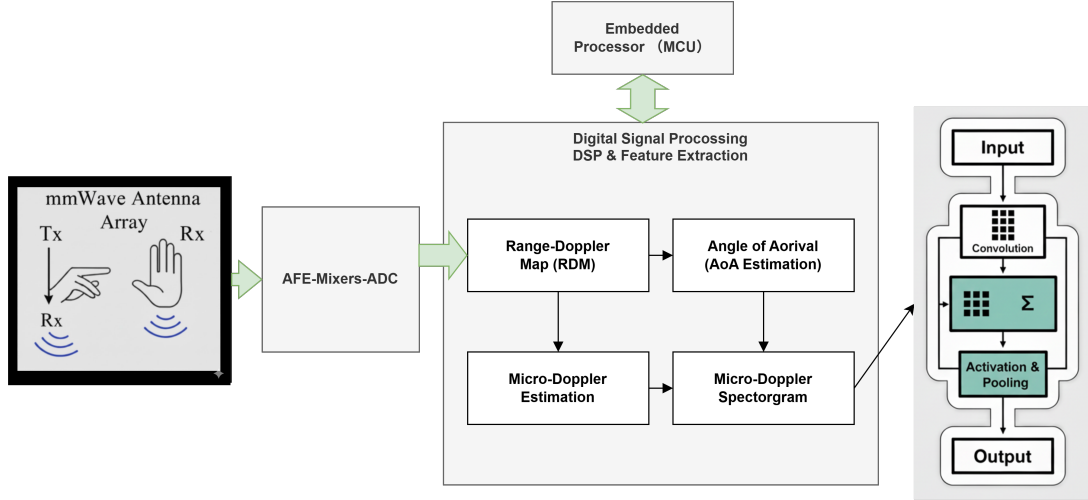


图 1.1 基于 Doppler 雷达的手势识别系统示意图

1.2.1 雷达数据采集

多普勒雷达手势识别的物理基石在于多普勒效应，即当波源与观察者之间存在相对运动时，观察者接收到的波频率发生改变的现象。在雷达手势感知中，这一效应被精细地用于刻画手部动力学特征。单频连续波（CW）雷达是实现多普勒检测最基础的体制。其发射信号通常表现为单频正弦波：

$$s_T(t) = A_T \cos(2\pi f_c t + \phi(t)) \quad (1.1)$$

其中， f_c 为载波频率， $\phi(t)$ 代表发射机振荡器的相位噪声。当电磁波照射到距离雷达 $d(t)$ 的手部目标并发生反射后，接收机获取的回波信号 $s_R(t)$ 经过往返时延 $\tau(t) = 2d(t)/c$ 产生相位偏移：

$$s_R(t) = A_R \cos(2\pi f_c(t - \tau(t)) + \phi(t - \tau(t))) \quad (1.2)$$

在接收前端，回波信号与发射信号的采样样本进行混频（Mixer）并经过低通滤波，产生基带信号。对于一个以径向速度 v 匀速运动的目标，其产生的时间变动相位项导出多普勒频移 f_D ：

$$f_D = -\frac{2v}{\lambda} = -\frac{2vf_c}{c} \quad (1.3)$$

公式表明，多普勒频移与目标的运动速度成正比，与载波波长成反比。由于毫米波雷达的波长极短（例如 60GHz 对应波长为 5mm），即使是微小的手指抖动（速度 v 极小）也能激发出显著的频率偏移，这奠定了其捕捉微细手势

(Micro-gestures) 的物理基础。

尽管单频 CW 雷达能精确测量速度，但其缺乏测距能力，且无法区分处于相同速度的多个目标。调频连续波 (FMCW) 体制通过对载波进行线性扫频解决了这一问题。其发射频率在周期 T_c 内随时间线性增加，带宽为 B 。回波信号与发射信号之间的差频信号（拍频信号）频率 f_{beat} 包含了目标的距离信息：

$$f_{beat} = \frac{2d}{c} \cdot \frac{B}{T_c} \quad (1.4)$$

通过对接收到的数字中频 (IF) 数据流执行二维 FFT 处理，可以生成距离-多普勒图 (Range-Doppler Map, RDM)。在 RDM 中，横轴通常代表目标的径向速度，纵轴代表距离，从而实现了在距离-速度平面的目标分离。此外，引入单输入多输出 (SIMO) 或多输入多输出 (MIMO) 天线阵列，利用各接收通道间的空间相位差，可以进一步解算出目标的方位角与俯仰角。多维感知能力的提升使得雷达不仅能识别手势的类别，还能实时重构手部在三维空间中的运动轨迹。

1.2.2 神经网络建模

收集得到的信号后，需要经过分类任务对应相应的动作行为，这一步通过一个神经网络 (Neural Network) 的建模实现。在多普勒雷达手势识别任务中，神经网络被建模为一个复杂的非线性映射函数 $f_\theta(\cdot)$ 。其核心目标是学习从高维雷达信号特征空间 $\mathcal{X}$ （如 Range-Doppler Map 或 Micro-Doppler Spectrogram）到离散手势标签空间 $\mathcal{Y}$ 的映射关系。

假设输入特征向量为 $\mathbf{x} \in \mathbb{R}^n$ ，一个具有 L 层的深度神经网络可以表示为一系列嵌套的非线性变换：

$$\hat{\mathbf{y}} = f_\theta(\mathbf{x}) = \sigma_L(\mathbf{W}_L \sigma_{L-1}(\dots \sigma_1(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) \dots) + \mathbf{b}_L) \quad (1.5)$$

其中， $\theta = \{\mathbf{W}_l, \mathbf{b}_l\}_{l=1}^L$ 代表网络的可学习参数集合， $\mathbf{W}_l$ 为权重矩阵， $\mathbf{b}_l$ 为偏置向量， $\sigma(\cdot)$ 为非线性激活函数（如 ReLU、Leaky ReLU 或 Sigmoid）。

神经网络的训练本质上是一个经验风险最小化 (Empirical Risk Minimization, ERM) 的优化过程。通过定义损失函数 $\mathcal{L}(\hat{\mathbf{y}}, \mathbf{y})$ 来衡量模型预测值与真实标签之间的偏差。在多分类手势识别任务中，通常采用交叉熵损失函数 (Cross-Entropy Loss)：

$$\mathcal{J}(\theta) = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K y_{i,k} \log(\hat{y}_{i,k}) + \frac{\lambda}{2} \|\theta\|_2^2 \quad (1.6)$$

其中 N 为样本总数, K 为手势类别数, λ 为正则化系数用于防止模型过拟合。参数更新过程遵循梯度下降法 (Gradient Descent):

$$\theta_{t+1} = \theta_t - \eta \cdot \nabla_{\theta} \mathcal{J}(\theta_t) \quad (1.7)$$

其中 η 为学习率, $\nabla_{\theta} \mathcal{J}(\theta_t)$ 为损失函数关于参数的偏导数, 通过反向传播算法^[10] (Backpropagation) 计算获得。

对于 K 类手势识别任务, 网络末端通常接入 Softmax 层, 将隐藏层的输出映射为各类别的概率预测分布 $P(y = k|\mathbf{x})$:

$$P(y = k|\mathbf{x}) = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}} \quad (1.8)$$

其中 z_k 是分类层对第 k 类的原始输出评分 (Logit)。最终的预测结果由最大后验概率 (MAP) 准则决定:

$$\text{Label} = \arg \max_{k \in \{1, \dots, K\}} P(y = k|\mathbf{x}) \quad (1.9)$$

在测试阶段, 模型参数 θ 被冻结, 不再进行更新。模型通过前向推断 (Inference) 对未见过的测试集 $\mathcal{D}_{test}$ 进行泛化能力评估。核心评价指标为分类准确率 (Accuracy):

$$\text{Accuracy} = \frac{1}{|\mathcal{D}_{test}|} \sum_{x_i \in \mathcal{D}_{test}} \mathbb{I}(f_{\theta}(x_i) = y_i) \quad (1.10)$$

其中 $\mathbb{I}(\cdot)$ 为指示函数, 当预测标签与真实标签一致时取值为 1, 否则为 0。

2 国内外研究现状

2.1 研究方向及进展

早期的雷达手势识别系统受限于硬件带宽与天线数量, 主要依赖于一维信号特征。主要关注幅度随时间的变化 (Time-Amplitude) 或幅度随距离的变化 (Range-Amplitude)。高分辨率距离像 (High-Resolution Range Profile, HRRP) 是这一阶段的核心^[11]。它通过宽频带信号处理, 获取目标在雷达射线方向上的散射点分布, 反映目标的径向结构特征。与此同时 1D 特征无法区分处于相同距离但不同方位角的目标 (如手部与躯干), 在复杂环境下极易产生目标混叠。

随着调频连续波 (FMCW) 技术的成熟, 2D 建模成为了手势识别的经典范式。通过对回波执行二维 FFT 处理, 将信号映射为距离-多普勒平面 (Range-Doppler

Map, RDM), 同时解算出目标的径向距离与径向速度。RDM 能够有效分离动态手部与静态背景杂波, 在此基础上, 通过短时傅里叶变换 (STFT) 提取出的微多普勒 (micro-Doppler) 指纹, 能够精细刻画手指屈伸等细微动作产生的频率调制^[12]。

4D 建模是目前最前沿的技术方向, 它标志着雷达从“探测传感器”向“高分辨率成像仪”的转变。维度定义: 4D 指的是距离 (Range)、径向速度 (Doppler/Velocity)、方位角 (Azimuth) 以及俯仰角 (Elevation/Height)。生成包含三维空间坐标与实时速度信息的密集点云 (Point Cloud), 每个点云点通常携带 (x, y, z, v, I) 等五个维度的数据, 其中 I 为反射强度。4D 建模引入了俯仰角信息, 在一定程度上解决了 3D 雷达无法区分“高处障碍物 (如桥梁)”与“路面目标 (如车辆)”的痛点。在手势识别中, 4D 建模支持对复杂 3D 运动轨迹 (如空中连续写字) 的高保真重构, 其点云密度与精度已接近低线束 LiDAR 性能。

全球雷达手势识别的研究热潮很大程度上源于 2016 年 Google 推出的 Project Soli。这一里程碑式的项目首次证明了利用 60 GHz 调频连续波 (FMCW) 雷达捕捉指尖微小动作的可能性。通过集成天线阵列与随机森林等机器学习算法, Soli 实现了对诸如“拨动转盘”、“摩擦手指”等复杂微动作的精准识别, 其追踪精度达到了亚毫米级别。此后, 国际半导体巨头如英飞凌 (Infineon)、德州仪器 (TI) 和意法半导体 (ST) 纷纷跟进, 推出了集成了硬件加速器 (HWA) 和微控制器 (MCU) 的毫米波雷达 SoC 芯片。

在学术界, 欧洲的研究机构如英飞凌、卡尔斯鲁厄理工学院 (KIT) 和比利时微电子研究中心 (imec) 在低功耗雷达架构和少样本学习 (Few-shot Learning) 领域占据前沿^{[13][14]}。例如, imec 致力于开发亚毫瓦级的脉冲雷达, 在 IEEE RFIC 2020 会议上展示了集成在标准 28nm CMOS 中的 60GHz 毫米波运动检测雷达。超灵敏雷达可实现 2 厘米的距离分辨率, 针对生命体征监测和手势识别进行了优化, 具有 5 米处心跳检测以及对行人位置和速度的精确跟踪的能力。紧凑型雷达芯片仅消耗 62mW^[15]。国际研究的重点已从早期的“单纯识别”转向“实时性、低功耗与泛化能力的平衡”, 强调在嵌入式平台上实现毫秒级的响应延迟。

此外, 国外研究针对标注雷达数据昂贵的问题, 研究人员开始利用扩散模型 (Diffusion Models) 生成高度逼真的合成多普勒指纹, 并结合物理建模缩短“仿真到现实”的差距 (Sim-to-Real Gap)^[16]; mmWrite 等系统利用大规模相控阵雷达实现毫米级的指尖追踪, 支持在任意平面上进行虚拟手写, 并结合双向 LSTM (Bi-LSTM) 进行字符识别, 识别准确度突破 98

2025 年第三季度发布的论文中提出在一个闭环系统中，雷达接收机对环境进行感知，学习目标与背景的相关信息，并根据预设任务目标自适应地调整发射机参数。尽管这一范式在频谱共享、目标检测与跟踪领域已取得显著进展，但在针对手势控制和智能环境交互的应用中，如何根据用户的即时状态调整雷达的物理层参数仍是一个相对空白的领域。Gurbuz et al. 等人首个将发射信号的自适应性与人类用户交互状态直接挂钩的研究工作。通过引入“人机在环”（Human-in-the-Loop）机制，雷达不再只是一个被动的数据采集器，而是一个能够理解交互上下文并主动优化其感知策略的智能实体^[17]。

中国在雷达手势识别领域展现出极强的后发优势，重点集中在算法层面的创新与特定场景的应用落地。《雷达学报》近期刊载了关于超宽带雷达人体动作四维成像数据集的研究，为国产雷达手势识别算法提供了核心基石；北京理工大学团队探讨了基于 Transformer 位置编码与序列编码的多帧 RDM 特征融合算法，强调了多维特征融合优于单特征识别的结论。

北京理工大学、南京邮电大学、清华大学以及电子科技大学等高校构成了研究的主力军。国内研究者敏锐地观察到雷达信号在高维特征利用上的冗余问题，提出了多种时空压缩特征表示学习方法。例如，南京邮电大学团队开发的毫米波雷达手势识别算法，通过对 FMCW 信号进行时空压缩特征提取，有效解决了深度学习模型在处理大规模原始雷达数据时的计算瓶颈^[18]。针对微小动作识别难的问题，国内学者提出了诸如 CQ-MobileNetV3 的轻量化网络结构，通过引入注意力机制（CBAM）和自注意力模块，在仅需 0.027 GFLOPs 计算量的情况下实现了极高的识别准确率^[19]。此外，国内研究在雷达与视觉融合、身份识别联合手势识别等方面也取得了重要进展，通过 Gram 匹配等损失函数优化，显著提升了跨场景的鲁棒性^[20]。

表2.1对比了近三年出现的几种轻量化手势识别网络的参数量、功耗性能和应用领域。可以看出在雷达采样之外，特定的神经网络和模型架构能够有效地提升识别任务的多样性，同时也会带来一定的计算资源负载。

2.2 存在问题

虽然基于毫米波的手势识别已经取得了非常多的研究进展与应用，其仍存在以下几个关键问题和挑战

- (1) **现实噪音扰动**：在实际复杂的室内人机交互场景中，目标手部并非处于孤立的真空中。人体躯干由于其巨大的反射面积（RCS），其回波强度通常比指

表 2.1 手势识别模型与系统性能对比

模型名称	核心架构	参数量	功耗/延迟	识别任务
CQ-Mobile NetV3	优化的 MobileNet + CBAM	0.207 M	27 MFLOPs	14 类微手势
TinyRadar NN ^[21]	2D CNN + TCN	46 K	120 mW	11 类通用手势
RadarNet ^[22]	Multi-head 2D CNN + LSTM	-	< 1 s	左右/上下扫动手势
No-ML System	纯 DSP (FFT + Azimuth)	N/A	< 10 ms	滑动、推压

尖运动回波高出 20-30 dB。这种极大的动态范围差异对 ADC 采样位数提出了严苛要求。更具挑战性的是，躯干的微小晃动或呼吸诱发的低频分量，在进行离散傅里叶变换（DFT）时，会由于窗函数的非周期性截断产生严重的频谱泄漏（Spectral Leakage），使得低频能量向高频区间扩散，从而像‘迷雾’一样掩盖了代表微手势特征的高频微弱多普勒分量。

- (2) **毫米波散射：**在 2021 年发表于 IEEE Transactions on Microwave Theory and Techniques 的关键论文中，Zhang 及其合作者指出，目前主流的微多普勒识别方法存在严重的鲁棒性缺陷。手部作为电磁波的散射体，其尺寸与毫米波波长处于同一数量级，属于复杂的 Mie 散射区^[2]。这意味着即使是同一个手势，由于手掌大小、手指曲率以及相对于雷达的入射角微小差异，回波产生的频谱纹理（Spectrogram）也会发生显著且不规则的变化，这导致不同照射角度、不同用户生理特征下的回波相位分布极不规则，极大地限制了传统线性重建算法的鲁棒性。
- (3) **神经网络算法限制：“黑盒”**深度学习方案具有样本依赖性：许多基于 CNN 或 RNN 的方法直接将时频图喂入模型。虽然识别率高，但通常需要数以万计的标记数据来覆盖不同用户的生理差异和执行风格。这对于个人便携式设备而言，训练成本过高，且模型泛化能力受限于数据集质量。并且基于深度学习的分类器通常在特定数据集上表现优秀，但在面对未见过的（untaught）新用户或新环境（如不同材质遮挡、不同距离/角度）时，识别准确率往往会大幅下降。
- (4) **存储和计算负载：**高维雷达数据（如 4D 点云或 RDM 序列）具有极大的冗余和计算负载。在内存和功耗受限的嵌入式 IoT 设备或移动终端上，运行复杂的 3D-CNN 或变体网络会面临严重的延迟问题。对于极高的雷达数据采样率（通常每秒数千个 Chirp），目前的主流做法是进行特征降维，例如将 3D 张量投影为距离-时间图（RTI）或多普勒-时间图（DTI）。然而，降维过

程不可避免地会导致空间信息的丢失。如何在极低功耗 MCU（如 Infineon PSoC Edge）上直接处理原始 IF 信号或保留更高维度的张量特征，是工业界亟需解决的硬件加速问题。

3 研究展望

针对上述问题，未来多普勒雷达手势识别的研究将朝着智能化、轻量化和多功能一体化方向演进，主要优化方向有以下几种：

- (1) **深度学习优化**：探索基于 MobileNetV3 等轻量化骨干网络与注意力机制（如 CBAM）的深度融合，在嵌入式平台上实现毫秒级响应。此外，利用脉冲神经网络（SNN）异步发射、事件驱动和高度稀疏化的特点，能够有效减少较大的训练参数且保持精度损失比较低。SNN 不再需要昂贵的乘累加（MAC）运算，而是采用简单的加法和门限发射机制，有望在神经形态芯片上实现比传统 DNN 低几个数量级的能耗，成为实现“常开式（Always-on）”感知的核心技术路径
- (2) **物理模型融入分类建模**：针对 Mie 散射带来的建模不确定性，研究将从纯数据驱动转向物理规律引导的分类范式。借鉴 Zhang et al. 的确定性信号分类思想，将手势运动学约束（如关节链运动矢量、RCS 动力学模型）引入神经网络损失函数中。通过结合高精度圆柱体网格（Cylinder Mesh）的手部物理建模与生成式 AI 技术，可以生成具备运动学一致性的合成多普勒谱图，有效桥接“仿真到现实”的差距（Sim-to-Real Gap），在数据稀缺的情况下提升模型的鲁棒性。
- (3) **元学习与数据快速适配**：引入 MAML^[23]（模型无关元学习）等框架，通过少量样本的快速微调，系统通过在大规模通用手势库中学习“如何学习”，构建具备快速迁移能力的初始化参数。在实际应用中，新用户仅需执行 1-5 次自定义样本（Few-shot），模型即可通过极少次的梯度微调自适应该用户的运动幅度和习惯，从而在智能手机、可穿戴设备中实现真正个性化的“开箱即用”体验。
- (4) **雷达大模型与 Foundation Model**：随着自监督预训练技术（如 RF-MAE）的成熟，针对雷达射频信号的通用大模型将成为行业底座。通过在数百万个未标记样本上学习通用的电磁表征，模型可以像视觉领域的大模型一样，通过简单的下游任务微调（Fine-tuning），同时完成手势识别、身份鉴权、跌倒检测及生命体征监测等多项任务。这种全场景感知能力将使雷达不仅是单一

的交互工具，更是未来智能家居生态中具备高度环境语义理解能力的“感知中枢”。

同时这种基于毫米波雷达建模的技术不仅可以适用于手势检测与交互，还可以用在临床护理与家庭康复、动态生物识别特征等领域中，传统的接触式监测设备往往由于导线束缚和皮肤刺激导致患者舒适度下降或者带来冗余的空间和资源占用。雷达生物探测的核心优势体现在：

- (1) **长期健康趋势追踪：**由于无需用户刻意佩戴，雷达可集成于床头柜或天花板，在睡眠状态下无感知地监测呼吸暂停（Sleep Apnea）事件及心率变异性（HRV）。
- (2) **身份识别与安全认证：**呼吸轨迹与心脏搏动的微多普勒分量具有唯一性。研究表明，基于呼吸动力学的高级特征（如吸气/呼气斜率）可实现高准确率的连续身份鉴权，且不会像生物特征图像那样引发泄露风险。
- (3) **极端场景的可行性：**对于早产儿、严重烧伤患者或传染病隔离区，无法使用接触式电极，雷达提供了唯一的连续监测方案，可以通过非接触式的高精度毫米波雷达实现安全隔离的体征监测。
- (4) **手势意图识别与用户身份验证的联合任务：**随着 IoT 设备的普及，用户对身份安全的关注日益增加，手势过程中蕴含了个人的生理特征（手掌大小、骨骼比例）和动作习惯（加速度模式），这可以作为一种动态生物识别特征用于“隐形鉴权”的授权检测。

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二、开题报告

1 问题提出的背景

1.1 背景介绍

在物联网（IoT）与泛在计算渗透社会各层级的当下，人机交互（HCI）正经历从“人适应机器”向“机器理解人”的本质演进。传统的触控交互与外设交互虽技术成熟，但在医疗手术室、车载驾驶舱及工业自动化等高标准场景中，物理接触不仅存在卫生风险，更受限於操作空间与视觉注意力的分配问题。随着 AR/VR 及智能座舱技术的普及，市场亟需一种高自由度、低认知负担且符合自然直觉的非接触式交互方案。手势作为人类最自然的表达载体，已成为实现“无感感知”交互的核心突破口。

当前手势识别的技术栈呈现三足鼎立之势，但在实际部署中面临各自的瓶颈：

- (1) **计算机视觉（CV）**：虽具高分辨率，但对光照高度敏感，且在暗光或遮挡下鲁棒性极差，更重要的是其在家庭、医疗等私密空间存在天然的伦理与隐私壁垒。
- (2) **穿戴式设备**：虽然定位精度极高，但设备带来的物理束缚感违背了“自然交互”的初衷，且长期佩戴的舒适度与维护成本限制了其大规模民用。
- (3) **射频传感（雷达）**以毫米波（mmWave）多普勒雷达为代表的射频感知，凭借其全天候工作特性、穿透遮挡的物理能力，以及仅获取运动特征而不获取图像信息的天然隐私保护优势，在智能终端领域展现出显著的竞争优势。

得益于半导体工艺向 28nm 及以下节点的迈进，毫米波雷达已实现超大规模集成（SoC）与微型化，Google Soli 项目的落地标志着该技术已具备进入消费电子市场的技术可行性。雷达利用多普勒效应捕获手指微小运动带来的频率偏移与相位调制。然而，在真实交互场景中，手部作为非刚体目标，其回波信号中蕴含的微多普勒特征极其微弱且易受躯干反射、环境噪声及多径效应的严重干扰。

因此，如何结合先进的信号处理技术（如 RDM 生成、AoA 估计）与深度学习模型，从复杂的原始信号中提取稳健的手势语义，实现低延迟、高精度的实时识别，是当前学术界与工业界共同关注的核心课题，也是本研究展开的基石。

1.2 本研究的意义和目的

本研究旨在针对当前非接触式人机交互中的瓶颈问题，基于毫米波多普勒雷达传感技术，构建一套高鲁棒性、高实时性的手势识别系统。本研究的核心目的是针对智能终端（如智能手机、平板电脑）的近距离交互场景，在基础上系统性实现并优化一种基于确定性多普勒信号分类的毫米波雷达手势识别系统。

首先，本研究旨在突破传统“点散射模型”在近场环境下的局限性。在菲涅尔近场区（Fresnel near-field range）内，手部不再能被简单视为一个质点，其复杂的生理结构和不规则的运动轨迹会导致严重的相位畸变^[1]。本研究通过引入米氏散射（Mie scattering）理论对手掌进行重新建模，旨在揭示手部精细动作与回波相位演变之间的确定性物理关联。

其次，本研究致力于实现一个端到端（End-to-End）的识别流水线（Pipeline）。这包括从底层的硬件配置、数据采集、模数转换（ADC），到中层的数字信号处理（DSP）以及高层的深度学习分类预测。通过这一流水线的建立，本研究将探索如何利用动态反射散射技术（DRS）提取鲁棒的手势特征，从而解决雷达感测中常见的强背景杂波掩盖微弱手势信号的问题。

最后，本研究拟解决轻量化模型在嵌入式端侧的实时推理问题。通过优化神经网络架构（如卷积层与全连接层的参数平衡），实现在资源受限的 SoC 平台上进行毫秒级响应，确保交互的流畅性。

本研究不仅具有深厚的学术探索价值，更具备迫切的现实应用意义，具体体现在以下三个维度：

- (1) **学术层面：**完善非刚体目标的近场感知理论现有的雷达识别研究多侧重于远距离的大动作分类，而本研究聚焦于近场微手势。通过将手掌视为米氏散射体，本研究为非刚体目标的电磁建模提供了新的视角。这有助于完善雷达信号处理中关于“确定性多普勒信号”的理论框架，为后续研究在复杂多径环境下提取高纯度运动特征提供参考依据。
- (2) **技术层面：**平衡识别精度、鲁棒性与计算负载传统深度学习方案具有严重的“数据饥渴”特性，且易受特定用户特征的影响。本研究通过物理建模引导的特征提取（DRS 技术），降低了模型对海量标注数据的依赖。这种“物理模型 + 深度学习”的结合方式，不仅能提升系统在真实噪音扰动下的鲁棒性，还显著降低了后端分类器的计算负荷，为毫米波雷达技术在移动终端的常态化部署铺平了道路。
- (3) **社会应用层面：**重塑自然、私密且安全的交互范式毫米波雷达具备“全天候、

非接触、非成像”的独特优势。相比于计算机视觉方案，雷达方案仅捕获运动包络而不获取图像信息，从物理层面上规避了隐私泄露风险。系统可在暗光、烟雾或存在视距遮挡的复杂场景下稳定工作。这种基于自然手势的交互方式，能有效解决医疗手术、车载驾驶等特定场景下物理接触带来的风险，极大地提升了人机交互的自由度与效率。

综上所述，本研究通过对前沿雷达感知框架的基础实现并优化，力求在复杂的物理世界与精准的数字指令之间建立一条稳健的感知链路，对于推动下一代自然用户界面（NUI）的发展具有重要的学术与应用参考价值。

2 论文的主要内容和技術路线

2.1 主要研究内容

拟设计并实现一套面向智能终端的毫米波多普勒雷达手势识别系统，主要研究工作从雷达信号建模、特征提取、深度学习分类器设计以及系统鲁棒性验证四个方面展开。本研究设计的系统囊括从物理世界数据采集、模数转换和分类预测结果的整个流水线（pipeline）。我们主要需要解决以下重要问题：

- (1) **数据传感的鲁棒性：**如文献综述中，Zhang et al. 团队提出：主流的毫米波微多普勒识别方法存在显著噪音扰动和手掌肌肤纹理散射的问题^[1]，我们将采用动态反射散射技术（DRS），将人类手掌视为米氏散射体（Mie scatter），而非传统的点源模型，并重点研究在菲涅尔近场区（Fresnel near-field range）内，如何通过确定性多普勒信号的分类来抵消环境杂波的影响，从而提升系统在智能设备近距离交互中的感知稳健性。
- (2) **数据传输的硬件层面支持：**设计低通滤波和 ADC 电路，并编写 MCU 的控制逻辑作为数据流的传输环节，将采样得到的物理世界信息加工处理，方便后续数据的分类和训练。
- (3) **面向智能终端的应用场景要求系统具备极低的响应延迟：**本研究拟解决如何在保证高识别率的前提下，对深度学习模型进行轻量化剪枝与算子优化。重点研究如何利用低复杂度的分类算法（如 SVM 或轻量化 CNN）处理经由 DRS 技术提取的确定性多普勒序列，使整个识别流水线（Pipeline）能在功耗受限的移动端 SoC 上实现实时闭环。

2.2 技术路线

2.2.1 雷达信号建模

对于第 i 个 TDM 接收通道，其数字基带 I/Q 信号可建模为

$$\begin{aligned} I_i[n] &= A_{I,i}[n] \cos(\Phi_i[n]) + dc_{I,i} + n_{I,i}[n], \\ Q_i[n] &= A_{Q,i}[n] \sin(\Phi_i[n]) + dc_{Q,i} + n_{Q,i}[n], \end{aligned} \quad (2.1)$$

其中 $A_{I,i}[n], A_{Q,i}[n]$ 表示随时间缓慢变化的幅度项, $dc_{I,i}, dc_{Q,i}$ 为直流偏置, $n_{I,i}[n], n_{Q,i}[n]$ 为噪声项, 而 $\Phi_i[n]$ 为多普勒相位。与幅度相比, 多普勒相位直接由电磁波传播路径长度变化决定, 因此是描述手势运动的关键信息。

离散时间下, 多普勒相位可以表示为相位增量的累积形式

$$\Phi_i[n] = \Phi_i[0] + \sum_{j=1}^n \Delta\Phi_i[j]. \quad (2.2)$$

为了从 I/Q 信号中稳定地提取相位增量 $\Delta\Phi_i[j]$, 并抑制幅度波动的影响, 采用反正弦 (arcsine) 解调算法, 其表达式为

$$\Delta\Phi_i[j] = \arcsin \frac{I_i[j]Q_i[j+1] - I_i[j+1]Q_i[j]}{\sqrt{I_i^2[j+1] + Q_i^2[j+1]}\sqrt{I_i^2[j] + Q_i^2[j]}}. \quad (2.3)$$

该公式利用归一化后的 I/Q 叉乘关系, 使提取的结果主要反映相位变化而非幅度变化, 从而得到与目标运动高度相关的相位序列。

在 5.8 GHz 频段下, 电磁波波长与人手尺寸处于同一量级, 手部散射属于 Mie 散射并发生在 Fresnel 近场区域, 不能简单视为点目标散射。为获得稳定且可重复的相位响应, 将手势规范为平面手掌, 并将其近似为一个导体矩形孔径。在 y 极化入射下, 孔径上感生的等效表面电流沿极化方向呈正弦分布, 可表示为

$$J(\zeta, \xi) = I_0 \sin \left[k \left(\frac{l}{2} - |\xi| \right) \right], \quad (2.4)$$

其中 (ζ, ξ) 为孔径局部坐标, l 为孔径沿极化方向的尺寸, k 为波数。孔径上的每一微元电流可等效为 y 极化赫兹偶极子, 其在方向 (r, θ, ϕ) 上产生的微分散射场为

$$dE_y = -\frac{jk\eta_0}{4\pi r} J(\zeta, \xi) e^{-jkr} (\cos^2 \theta \sin^2 \phi + \cos^2 \phi) d\zeta d\xi. \quad (2.5)$$

接收天线处的总散射电场由对整个孔径的积分得到

$$E_R = \int_{-l/2}^{l/2} \int_{-w/2}^{w/2} G_R(\theta_R, \phi_R) dE_y, \quad (2.6)$$

其中 $G_R(\theta_R, \phi_R)$ 为接收天线方向图。该近场散射模型表明, 对于沿 $\pm x$ 、 $\pm y$ 或 $\pm z$ 方向的规范手势, 接收信号的相位响应在主瓣范围内呈现近似平坦特性, 从而使相位随手部位移变化表现为准线性且可重复的函数形式, 这是后续“确定性多普勒相位”的物理基础。

在几何层面, SIMO 多普勒雷达系统可等效为多个共享同一发射端的双基地雷达。设目标在时刻 t 的位置为 (x_S, y_S, z_S) , 则双基地往返距离为

$$r(t) = |\mathbf{r}_T(t)| + |\mathbf{r}_R(t)|, \quad (2.7)$$

其中 $\mathbf{r}_T(t)$ 和 $\mathbf{r}_R(t)$ 分别表示目标到发射端和接收端的距离向量。若目标以速度 $\mathbf{v}(t)$ 运动, 则往返距离的时间导数可写为

$$\frac{dr(t)}{dt} = \mathbf{v}(t) \cdot \hat{\mathbf{r}}_T - \mathbf{v}(t) \cdot \hat{\mathbf{r}}_R = |v_T(t)| - |v_R(t)|. \quad (2.8)$$

根据多普勒效应, 瞬时多普勒频移为

$$f_d(t) = \frac{|v_R(t)| - |v_T(t)|}{c} f_c, \quad (2.9)$$

其中 f_c 为载频, c 为光速。对多普勒频移随时间积分即可得到多普勒相位, 由此可建立路径长度与相位之间的等价关系

$$r(t) = -\frac{c}{2\pi f_c} \Phi(t). \quad (2.10)$$

该关系清晰地表明, 多普勒相位 $\Phi(t)$ 本质上编码了目标往返距离的变化过程, 从而直接反映了手势的几何运动轨迹。

在 SIMO 架构下, 不同接收天线对应不同的双基地几何关系, 因此同一手势在各通道中会产生形态相似但时间位置不同的相位曲线 $\Phi_i(t)$ 。对于规范定义的单向手势, 实验与理论分析均表明 $-\Phi_i(t)$ 通常呈凸形, 并且仅包含一个显著的最小值。为消除手势执行速度的影响, 将每段手势在时间轴上归一化到区间 $(0, 1)$, 并对 $-\Phi_i(t)$ 进行多项式拟合以抑制噪声, 然后提取每个通道相位最小值

出现的归一化时刻 $t_{m,i}$ ，构造特征向量

$$\mathbf{t}_m = (t_{m,1}, t_{m,2}, \dots, t_{m,M}). \quad (2.11)$$

该特征向量刻画了不同接收通道之间相位响应的相对时序结构，对手势高度、速度以及个体差异具有较强鲁棒性。

综上所述，通过

$$(I/Q) \rightarrow \Phi_i(t) \rightarrow \text{多通道几何差异} \rightarrow \mathbf{t}_m \quad (2.12)$$

的推理链条，将高维连续时间信号压缩为低维、可解释且物理意义明确的确定性特征，从而能够利用轻量级分类器实现高精度、低样本复杂度的手势识别。

2.2.2 神经网络建模

在雷达手势识别任务中，可将“手势”视为一个监督学习的多分类问题：给定某一时间窗口内的雷达观测序列（由基带 I/Q 或由相位序列派生的特征），预测其对应的手势类别标签。设手势类别集合为 $\mathcal{C} = \{1, 2, \dots, C\}$ ，训练数据集为

$$\mathcal{D} = \{(\mathbf{x}^{(n)}, y^{(n)})\}_{n=1}^N, \quad y^{(n)} \in \mathcal{C}. \quad (2.13)$$

其中 $\mathbf{x}$ 为输入特征，可由雷达信号处理模块输出。针对本文“确定性多普勒”建模思路，常用的神经网络输入有两类：低维确定性特征（最小值时刻向量）与高维时频图（谱图）。

若采用“确定性多通道最小值时刻”作为输入，则对 M 个接收通道，在对每段手势进行时间归一化与平滑拟合后，可提取每个通道的相位最小值出现时刻 $t_{m,i} \in (0, 1)$ ，并组成向量

$$\mathbf{x} = \mathbf{t}_m = (t_{m,1}, t_{m,2}, \dots, t_{m,M})^\top \in \mathbb{R}^M. \quad (2.14)$$

此时可用一个多层感知机（MLP）作为分类器 $f_\theta: \mathbb{R}^M \rightarrow \mathbb{R}^C$ ，其前向传播可写为

$$\mathbf{h}_0 = \mathbf{x}, \quad \mathbf{h}_\ell = \sigma(\mathbf{W}_\ell \mathbf{h}_{\ell-1} + \mathbf{b}_\ell) \quad (\ell = 1, \dots, L), \quad (2.15)$$

$$\mathbf{z} = \mathbf{W}_{L+1} \mathbf{h}_L + \mathbf{b}_{L+1} \in \mathbb{R}^C, \quad (2.16)$$

其中 $\sigma(\cdot)$ 为非线性激活函数（如 ReLU / GELU）， $\theta = \{\mathbf{W}_\ell, \mathbf{b}_\ell\}$ 为可学习参

数, $\mathbf{z}$ 为 logits。预测分布由 softmax 给出:

$$p_{\theta}(y = c | \mathbf{x}) = \frac{\exp(z_c)}{\sum_{k=1}^C \exp(z_k)}. \quad (2.17)$$

训练目标采用多分类交叉熵 (可带正则化):

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{n=1}^N \left(-\log p_{\theta}(y^{(n)} | \mathbf{x}^{(n)}) \right) + \lambda \|\theta\|_2^2. \quad (2.18)$$

推理阶段输出类别为

$$\hat{y} = \arg \max_{c \in \mathcal{C}} p_{\theta}(y = c | \mathbf{x}). \quad (2.19)$$

由于输入维度 M 很小 (例如 $M = 3$), 该 MLP 本质上是一个 “用神经网络替代线性/核方法分类器” 的建模方式, 优点是实现简单、训练开销小、对小样本更友好, 且与确定性特征的物理可解释性相一致。

若采用时频谱图作为输入, 则需先从多通道相位序列 $\Phi_i[n]$ 计算短时傅里叶变换 (STFT) 并构造谱图。对第 i 个通道, 设窗函数为 $w[\cdot]$ 、帧移为 H , 则谱图可表示为

$$S_i[m, \omega] = \sum_n \Phi_i[n] w[n - mH] e^{-j\omega n}, \quad X_i[m, k] = |S_i[m, \omega_k]|^2. \quad (2.20)$$

将 M 个通道的谱图堆叠为张量输入 (可视为多通道 “图像”):

$$\mathbf{X} \in \mathbb{R}^{M \times T \times F}, \quad (2.21)$$

其中 T 为时间帧数、 F 为频点数。然后使用卷积神经网络 (CNN) 分类器 $g_{\theta} : \mathbb{R}^{M \times T \times F} \rightarrow \mathbb{R}^C$:

$$\mathbf{U}_0 = \mathbf{X}, \quad \mathbf{U}_{\ell} = \sigma(\text{BN}(\text{Conv}_{\ell}(\mathbf{U}_{\ell-1}))) \quad (\ell = 1, \dots, L), \quad (2.22)$$

$$\mathbf{v} = \text{Pool}(\mathbf{U}_L), \quad \mathbf{z} = \mathbf{W}\mathbf{v} + \mathbf{b} \in \mathbb{R}^C, \quad (2.23)$$

并同样通过 softmax 得到 $p_{\theta}(y | \mathbf{X})$, 用交叉熵最小化

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{n=1}^N \left(-\log p_{\theta}(y^{(n)} | \mathbf{X}^{(n)}) \right) + \lambda \|\theta\|_2^2. \quad (2.24)$$

该 CNN 建模能够直接从高维谱图中学习判别模式，但通常需要更大数据量与更高训练成本；相比之下，基于 t_m 的低维 MLP 更贴合“确定性多普勒特征”的小样本设定。

为增强跨被试泛化能力，可引入特征标准化与数据增强。对低维特征可做

$$\tilde{x} = \frac{x - \mu}{\sigma}, \quad (2.25)$$

对谱图可做幅度归一化、随机时间裁剪、随机噪声扰动等。训练时可采用小批量优化（Adam^[2]/SGD^[3]）：

$$\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}(\theta), \quad (2.26)$$

其中 η 为学习率。最终，神经网络模型完成从“雷达观测”到“手势类别”的端到端映射，并可根据输入特征选择低维 MLP（确定性特征）或 CNN（谱图特征）两种实现路径。

2.3 可行性分析

调频连续波（FMCW）雷达在测距与测速领域的物理机制已极其成熟。特别是 Zhang et al. 团队提出的动态反射散射（DRS）技术^[1]，为近场环境下的非刚体手势建模提供了坚实的数学框架。通过将手掌定义为米氏散射体（Mie scatterer），能够有效解释近场菲涅尔区内的相位调制规律。前期对确定性多普勒信号（Deterministic Doppler Signals）的推导证明，通过捕捉相位的导数关系，可以稳定提取出手势的特征向量，这为本课题的在经典确定性多普勒建模框架基础上进行系统化实现与工程优化工作提供了核心理论指引。

与此同时地，当前半导体工艺的跨越式发展为本研究提供了可靠的硬件载体。以英飞凌（Infineon）BGT60TR13C 为代表的 60GHz 毫米波雷达 SoC 已经实现了超大规模集成，能够提供高达 4GHz 的扫频带宽，确保了厘米级的距离分辨率；现代嵌入式微控制器（如 Cortex-M55 核心）集成了高性能 ADC 模块与硬件加速器，采样率可达 40kHz 以上，完全满足手势信号的 Nyquist 采样需求。

在算法层面，轻量化深度学习与信号处理库的普及大幅降低了开发门槛：现有的数字信号处理（DSP）框架可以高效实现 2D-FFT、STFT 以及 MTI 滤波等预处理步骤，从而快速生成高质量的距离-多普勒图（RDM）和微多普勒谱图。借助 TensorFlow 或 PyTorch 等成熟工具，能够快速构建并训练轻量化神经网络（如结合了 CBAM 注意力机制的模型），实现对手势语义的端到端识别。Zhang 等人

的研究已给出明确的参数配置建议（如 32 帧/秒的帧率、线性化 Chirp 设计等），这为本研究在初期调试阶段规避系统性错误、快速进入算法优化阶段提供了宝贵的工程参考。

3 研究计划进度安排及预期目标

3.1 进度安排

在该研究项目中我们的时间规划与安排如表3.1所示。

表 3.1 研究进度安排与时间规划

时间区间	研究阶段	关键任务与预期成果
2025.12 -- 2026.01.10	文献调研与开题报告	分析 Zhang et al. 论文核心架构，完成基于确定性多普勒信号分类的开题论证。
2026.01.11 -- 2026.02.15	物理建模与数值仿真	构建菲涅尔近场区手部 Mie 散射模型，利用仿真获取 DRS 特征模式。
2026.02.16 -- 2026.03.10	硬件链路和数据采集	配置雷达 $F_s > 40\text{kHz}$ 采样率链路，构建包含非刚体微动特征的实验数据库。
2026.03.11 -- 2026.04.10	算法优化与系统部署	训练轻量化神经网络模型，完成全流水线在嵌入式智能终端 SoC 的移植与实时测试。
2026.04.11 -- 2026.05.20	实验评估与论文撰写	进行跨场景鲁棒性实验，对比射频方案与视觉方案，完成毕业论文撰写及答辩工作。

3.2 预期目标

本研究预期通过对毫米波雷达手势识别全流水线的基础改进与优化，达成以下具体目标：

- (1) 预期建立一套基于米氏散射（Mie scattering）理论的手部电磁模型，准确描述在菲涅尔近场区内手掌运动产生的确定性多普勒相位调制规律。通过物理仿真验证动态反射散射（DRS）特征的有效性，为消除环境多径干扰提供物理层依据。
- (2) 预期完成一套集成 ADC 采样、数字信号处理及深度学习预测的实时手势识别系统。系统需在硬件链路端实现高于 40kHz 的稳定采样，并提出在算法端实现对“滑动”、“推压”等典型微手势的精度分类。目标是将系统端到端

响应延迟控制在 200ms 以内，满足实时交互的需求。

- (3) 预期证明该系统在暗光环境、视距 (LoS) 部分遮挡及存在显示器电磁噪声干扰的情况下，识别准确率显著优于传统视觉方案。最终将轻量化神经网络成功部署于嵌入式 SoC 平台，验证其在移动终端侧低功耗运行的可行性，为下一代无感、私密的人机交互接口提供技术范式。

4 参考文献

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三、外文翻译

通过分类确定性多普勒信号实现智能设备的手势识别

1 摘要

智能手机和平板电脑等个人设备正迅速成为个人通信、信息和控制中心。除了多点触控屏幕之外，人类手势被视为一种新型的人与智能设备交互界面。在本研究中，我们提出了一种非接触式解决方案，用于在智能设备上实现手势识别。该方案基于连续波、时分复用（*TDM*）、单输入多输出（*SIMO*）的多普勒雷达传感器，可通过对现有智能设备射频前端进行轻微修改来实现；同时结合机器学习算法，通过对确定性多普勒信号进行分类来识别预定义的手势。我们搭建了一个模拟基于智能手机雷达传感器的实验平台，实验结果验证了所提方法的鲁棒性和准确性。关键词——分类，多普勒效应，手势识别（*HGR*），机器学习，雷达架构。

2 引言

近年来，智能手机和平板电脑等个人设备已成为人类日常生活中不可或缺的工具。这些智能设备集成了无线接口、传感器和强大的计算能力，正迅速成为个人通信、信息和控制中心 [1]。迄今为止，人与智能设备之间的大多数交互仍基于多点触控屏幕 [2]。随着智能设备的应用扩展到虚拟现实以及电视（*TV*）和投影仪的无线控制等场景，基于人体手势识别（*HGR*）的非接触式交互吸引了越来越多的关注 [3]–[6]。

最初，非接触式手部手势识别（*HGR*）在计算机科学领域被视为一种人机界面（*HMI*） [7]。基于摄像头和数字图像处理算法，这种无线 *HMI* 可归类为光学方法。由于高度依赖计算资源、直接视线以及照明条件，基于计算机视觉的 *HGR* 应用至今仍受限 [8]。为了克服上述成本、障碍物和光照问题，研究人员已开始探索基于微波和毫米波的方法，即脉冲和连续波（*CW*）雷达方法的 *HGR* [9]–[18]。

迄今为止，大多数已报道的基于 *HGR* 均采用调频连续波（*FMCW*）方案 [12]–[15], [19], [20]。其中最著名的是“*Google Soli*” [15], 它是一个 60 *GHz* 的 *FMCW* 雷达模块，带宽高达 7 *GHz*。通过分类距离-多普勒频谱图 [10], 它可以识别细微的手势，例如手指摩擦动作，可用于控制智能手机的音量或音乐播放列

表等功能。然而，额外增加的毫米波雷达芯片及其外围组件会提高成本，增加功耗，并占用更多内部空间，而这些资源对于空间和资源受限的设备来说十分稀缺。

到目前为止，在上述基于雷达的解决方案中已提出了两类识别算法。其中一类基于线性方法，试图从后向散射场 [4]-[6], [9] 中提取目标的位移信息，即手势的运动模式。这类方法的优点是所提取的手势模式具有明确意义，能与实际手势直接对应。缺点是由于手部的米氏散射 [21]，检测到的模式形状会因照明角度不同而产生显著变化，从而增加了识别难度。

另一类算法基于人工智能 (AI)，利用人工神经网络 (ANN) 和基于机器学习的分类方法的非线性泛化能力 [22]-[24]。这些方法用于识别手势散射场的“特征签名”。其优点在于，只要使用这些签名来训练基于 ANN 的分类器 [10], [12]-[14], [16], [17], [19], [25]，就可以对任意预定义的手势实现“黑箱式”识别。对于此类黑箱分类，需要大量来自不同人员完成的手势数据以训练分类器。然而，所需训练手势的数量可能高达数万，导致在手势数据集采集、数据标注和数据清洗方面耗费大量人力和时间。

在本研究中，我们结合了上述架构和算法的优点，提出了一种在智能设备上实现手部手势识别 (HGR) 的系统性解决方案。该方案基于连续波 (CW)、时分复用 (TDM)、单输入多输出 (SIMO) 的多普勒雷达传感器 (DRS) 以及一种通过分类确定性多普勒信号来识别手势的机器学习算法。这种 DRS 只需对现有智能设备的射频前端进行轻微修改即可实现，硬件成本、功耗和空间占用的增加可忽略不计。与“黑盒”分类不同，此类确定性多普勒信号具有明确的物理和数学意义，因此仅需使用单人完成的少量手势即可训练出分类器，并基于此获得对不同用户均具有高精度的手势识别能力。基于模拟智能手机 DRS 系统的实验验证了所提出 HGR 方法的鲁棒性和准确性。

本文其余部分组织如下。在第二部分中，回顾了相关现有的 DRS 研究工作，并在此基础上介绍了基于智能手机的 TDM SIMO DRS 架构以及专用于该 DRS 的算法。在第三部分中，推导了所提出 DRS 的手势设计及相应算法。第四和第五部分分别提供了全波仿真和实验验证。最后，第六部分给出了结论。

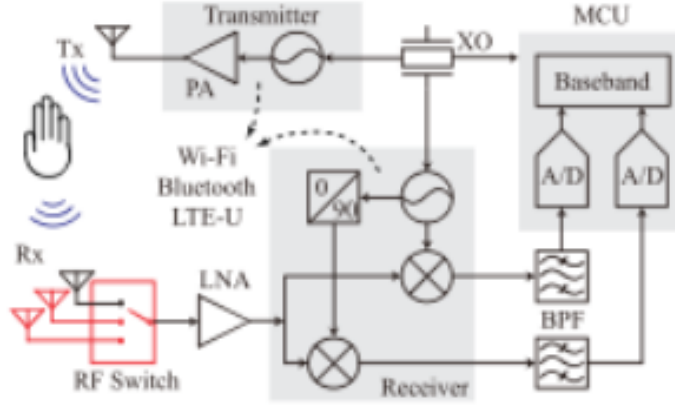


图 1 通过重构 Wi-Fi 和 LTE-URF 信道实现 TDM-SIMO DRS。红色绘制的设备为新增设备。

3 架构

3.1 架构

随着射频集成电路 (RFIC) 的快速发展,在过去十年中已开发出小型化的连续波 *DRS*,用于非接触式运动检测。目标运动包括人体生命体征 [26]-[30] 和机械运动 [31]-[33]。最初使用的是单输入单输出 (*SISO*) 架构的连续波 *DRS* [26], [27], [29], [30]。在能够消除相位模糊的算法辅助下,可实现对一维运动的线性检测。对于二维运动,也实现了具有一路发射和两路接收通道的 *SIMO DRS* [4], [36], [37]。然而,在上述所有 *SIMO DRS* 中,发射和接收天线被布置在一条直线上,因此无法用于检测三维手势。此外,由于包含多个射频通道,它们也难以集成到智能设备中。

在本研究中,利用连续波差分接收信号 (*CW DRS*) 的超窄带特性,我们提出了一种时分复用-单发多收 (*TDM-SIMO*) *DRS*,该方案只需对传统智能设备中的现有射频前端进行轻微修改即可实现。其示意图如图 1 所示。

迄今为止,智能设备尤其是智能手机和平板电脑已经集成了多种无线接口。除了用于 2G、3G、4G 和 5G 通信的射频前端外,还有工作在 2.4 GHz 和 5.8 GHz 频段的 *Wi-Fi*、蓝牙以及长期演进 (*LTE*) -U [38] 信道。在大多数设备中,所有这些信道均采用共享同一参考振荡器和基带单元的标准数字中频架构实现。因此,这些信道可灵活选择用于实现适用于 *HGR* 的 2.4 GHz 或 5.8 GHz 前端。

例如,可以将 2.4 GHz 蓝牙的发射信道与 2.4 GHz *Wi-Fi* 的接收信道结合,从而构成一个 2.4 GHz 前端。类似地,可以将 *Wi-Fi* 和 *LTE-U* 的 5.8 GHz 发射与接收信道结合,以获得适用于高分辨率感知 (*HGRs*) 的 5.8 GHz 前端。此

类组合可通过在基带重新配置相关信道来实现，无需任何硬件更改。

为了实现三维手势的高分辨率感知 (*HGRs*)，如后文所分析，需要一个具有一个发射信道和至少三个接收信道的单发多收 (*SIMO*) *DRS*。在本研究中，通过对接收信道进行复用实现该目标。如图 1 所示，采用一个单刀三掷 (*SP3T*) 射频开关连接三个接收天线，实现三个时分复用接收信道。需要注意的是，这种复用不会影响 *Wi-Fi*、蓝牙或 *LTE-U* 的正常运行。由于人体手势是秒级速度的运动，即使 10 Sa/s 的采样率也已足够。例如，对于一个采样频率为 1 MHz 的模数转换器 (*ADC*)，完成连续三次采样在每个 10 毫秒 周期内仅占用几微秒的时间片。

根据上述讨论，*HGR* 前端可以通过重新配置智能设备现有的射频通道来实现。唯一需要增加的器件是一个 *1P3T* 开关和两根额外的天线。

3.2 检测信号

对于图 1 所示的正交接收机，由第 i 个时分复用通道数字化的基带 I/Q 信号可表示为

$$\begin{aligned} I_i[n] &= A_{I,i}[n] \cos(\Phi_i[n]) + dc_{I,i}[n] + n_{I,i}[n] \\ Q_i[n] &= A_{Q,i}[n] \sin(\Phi_i[n]) + dc_{Q,i}[n] + n_{Q,i}[n] \end{aligned} \quad (1)$$

其中 $A_{I,i}$ 和 $A_{Q,i}$ 分别为 I/Q 信号的幅度， $dc_{I,i}$ 和 $dc_{Q,i}$ 分别为两个通道的直流偏移， $n_{I,i}$ 和 $n_{Q,i}$ 为加性噪声。直流偏移 $dc_{I,i}$ 和 $dc_{Q,i}$ 可通过文献 [39] 中提出的梯度下降算法消除。在以下讨论中，我们将省略单通道分析中的通道索引 i 。

在基带上，离散多普勒相位可以表示为

$$\Phi[n] = \Phi[0] + \sum_{j=1}^n \Delta\Phi^j \quad (2)$$

其中 $\Delta\Phi^j$ 是 j 次采样的增量多普勒相位。此前，已提出一些旨在线性提取多普勒信号的算法，例如扩展微分与交叉相乘 (*DACM*) [35] 以及反正弦算法 [4]。在本研究中，将使用反正弦算法来提取 $\Delta\Phi^j$ ，其为

$$\Delta\Phi^j = \arcsin \frac{I[j]Q[j+1] - I[j+1]Q[j]}{\sqrt{I^2[j+1] + Q^2[j+1]} \cdot \sqrt{I^2[j] + Q^2[j]}} \quad (3)$$

4 理论

4.1 手势的定义

图 2 显示了一部放置在 xy 平面上的智能手机。假设图 1 中所示的天线 T 、 R_1 、 R_2 和 R_3 位于智能手机的侧面。通常，此类天线为倒 F 型天线，极化方向沿 y 方向。

在本研究中，定义了用于实现多点触控屏幕主要功能的手势，即在主界面与子界面之间切换、上下滚动页面以及点击操作。对于图 2 中定义的 x - y - z 坐标系，上述手势分别对应于手在智能手机屏幕上方沿 $\pm x$ 、 $\pm y$ 和 $\pm z$ 方向的移动。它们分别被称为“左”(L)、“右”(R)、“前”(F)、“后”(B) 和“点击”(TAP) 手势。

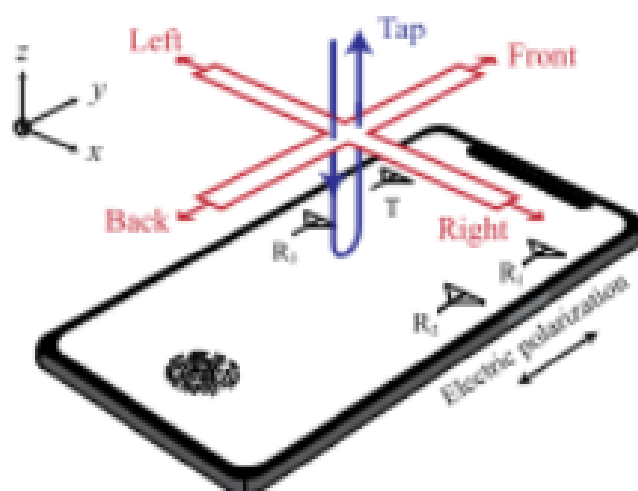


图 2 手势的定义

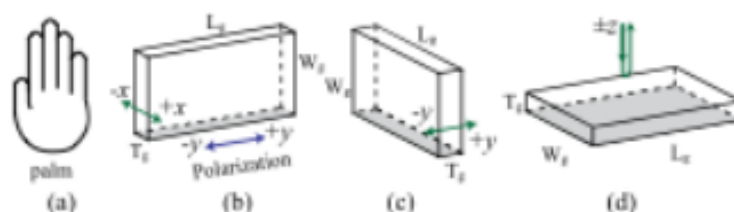


图 3 在 y 极化电场中对手势的调控。(a) 手掌。(b) $\pm x$ 手势。(c) $\pm y$ 手势。(d) $\pm z$ 手势。

4.2 错误矫正机制

在实际应用中，水平手势很容易偏离 x -轴和 y -轴，从而导致潜在的错误判断。为解决此问题，定义了两种冗余手势，即沿对角线方向的手部运动，即 $y = x$ 和 $y = -x$ ，以最大化容错能力。在实际操作中，一旦识别出手势为对角线运动，

则该手势将被判定为无效手势。此类冗余手势消除了 L 、 R 、 F 和 B 手势之间的模糊区域，分别记为 LF 、 LB 、 RF 和 RB 。

4.3 菲涅尔近场中移动手掌的散射

在本研究中，使用了一个 5.8 GHz 的 DRS 来演示此类手势的识别。在此频率下，对应的波长为 5.2 cm，与人手的尺寸相当。在这种情况下，人手掌应被视为米氏散射体，且分析应限于菲涅尔近场范围 [40]。

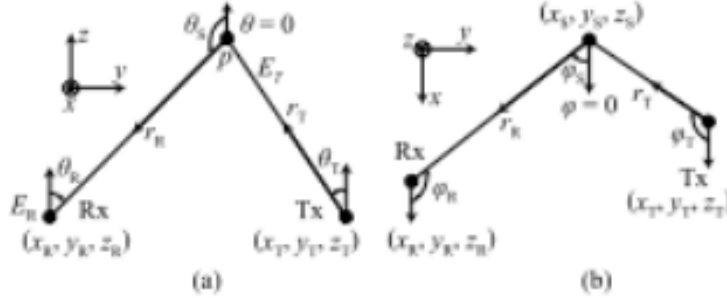


图4 E 移动手掌散射分析的双基地模型。(a) yOz 视图。(b) xOy 视图。

为了简化对人体手部米氏散射的分析，我们规定手势将通过平展的手掌执行，如图 3(a) 所示。然后，该手掌可近似为一个导电的扁平长方体，以不同的方向在 $\pm x$ 、 $\pm y$ 和 $\pm z$ 方向上移动。在图 3(b)-(d) 中，长方体的长度、宽度和厚度分别用 L_g 、 W_g 和 T_g 表示。阴影面为面向发射和接收天线的一面。当手掌在水平和垂直方向移动时，长方体的底侧和前面会将大部分入射场散射回接收天线。

在天线理论中，连续表面的散射可被视为来自孔径天线的二次辐射。通常，远场方程（如 *Friis* 方程）以及线天线的辐射方向图可方便地用于分析此类散射。然而，这些方法在菲涅尔近场中并不适用。

此处采用一种 *ab initio* 方法，通过沿矩形孔径的 x - 和 y - 方向积分 y 极化赫兹偶极子的辐射，来分析此类近场散射。在图 4 所示的全局 x - y - z 坐标系统中，假设发射天线和接收天线的相位中心分别位于 (x_T, y_T, z_T) 和 (x_R, y_R, z_R) 。垂直轴指向 $+z$ 方向。我们还假设散射孔径的几何中心位于 (x_S, y_S, z_S) ，其垂直轴指向 $+z$ 方向。

从数学上讲，发射天线照射到散射孔径上的电场可表示为

$$\mathbf{E}_T(r_T, \theta_T, \varphi_T) = \hat{\mathbf{g}} E_0 G_T(\theta_T, \varphi_T) e^{-jk r_T} / r_T \quad (1)$$

其中 $G_T(\theta_T, \varphi_T)$ 是发射天线的辐射方向图， r_T 是从发射天线到该单元的距离。类似地，我们将 $G_R(\theta_R, \varphi_R)$ 定义为接收天线的辐射方向图， r_R 为从该单元到接

收天线的距离。这里可以使用远场辐射方向图，因为智能设备中使用的四分之一波长倒 F 天线的远场菲涅尔范围较小。

根据图 4 中定义的坐标系，对于散射孔径上偏离 (x_S, y_S, z_S) 、记为 $(x_S + \zeta, y_S + \xi, z_S)$ 的点，所有坐标系中的仰角、方位角以及距离可以确定为

$$\left\{ \begin{array}{l} \theta_T = \arctan(\sqrt{(x_S + \zeta - x_T)^2 + (y_S + \xi - y_T)^2} / (z_S - z_T)) \\ \varphi_T = \arctan[(y_S + \xi - y_T) / (x_S + \zeta - x_T)] \\ \theta_S = \arctan(\sqrt{(x_R - x_S - \zeta)^2 + (y_R - y_S - \xi)^2} / (z_R - z_S)) \\ \varphi_S = \arctan[(y_R - y_S - \xi) / (x_R - x_S - \zeta)] \\ \theta_R = \arctan(\sqrt{(x_S + \zeta - x_R)^2 + (y_S + \xi - y_R)^2} / (z_S - z_R)) \\ \varphi_R = \arctan[(y_S + \xi - y_R) / (x_S + \zeta - x_R)] \\ r_T = \sqrt{(x_S + \zeta - x_T)^2 + (y_S + \xi - y_T)^2 + (z_S - z_T)^2} \\ r_R = \sqrt{(x_S + \zeta - x_R)^2 + (y_S + \xi - y_R)^2 + (z_S - z_R)^2} \end{array} \right. \quad (2)$$

在 y 极化入射光照射下，矩形散射孔径上感应的电流在 y 方向上满足正弦分布 [40]。它由沿 x 方向的无限长线电流组成，每个线电流具有如下表面电流分布：

$$J(\zeta, \xi) = I_0 \sin \left[k \left(\frac{l}{2} - |\xi| \right) \right] \quad (3)$$

其中幅度 $I_0 = \sigma_e E_T(r_T, \theta_T, \varphi_T)|_{\zeta=0, \xi=0}$ ， σ_e 为等效电导率。该电流可进一步划分为无限多个 y 极化的赫兹偶极子。对于位于 $(x_S + \zeta, y_S + \xi, z_S)$ 的赫兹偶极子，其 y 分量远场辐射可推导为（见附录）

$$\begin{aligned} dE_S^y(\zeta, \xi, r, \theta, \varphi) \\ = \frac{-jk\eta_0 J(\zeta, \xi) e^{-jkr}}{4\pi r} (\cos^2 \theta \sin^2 \varphi + \cos^2 \varphi) d\zeta d\xi. \end{aligned} \quad (4)$$

然后，通过沿 x - 和 y 方向对所有此类赫兹偶极子的辐射进行积分，可得到矩形孔径散射的总场，接收天线检测到的电场可推导为

$$E_R = \int_{-l/2}^{l/2} \int_{-w/2}^{w/2} G_R(\theta_R, \varphi_R) \cdot dE_S^y(\zeta, \xi, r_R, \theta_R, \varphi_R). \quad (5)$$

图 5(a)-(c) 展示了基于 (6) 式在 20 cm 的近场菲涅尔距离处计算得到的随 θ 变化的幅度和相位辐射方向图，分别对应“左/右”、“后/前”和“点击”手势，其

中 L_g 、 W_g 和 T_g 分别为 15、8 和 2 cm。

可以看出，所有幅度图样中的主瓣都较宽，有利于覆盖所有接收天线。在图 5(a) 中，由于 x （或 $\varphi = 0/180^\circ, \theta = 90^\circ$ ）方向上的 y 极化照射，其相位偏移几乎保持恒定，且不受“左/右”移动的影响。类似平坦的相位偏移区域也可在图 5(b) 和 (c) 中分别对应于 $\pm y$ （或 $\varphi = 90^\circ/-90^\circ, \theta = 90^\circ$ ）和 $\pm z$ （或 $\theta = 0/180^\circ$ ）方向上找到，它们分别适用于“前后”和“轻触”手势。如下文所示，这种平坦的相位响应将产生准线性、确定性的多普勒相位，这对 HGR 至关重要。上述手势的定义也正是基于此。

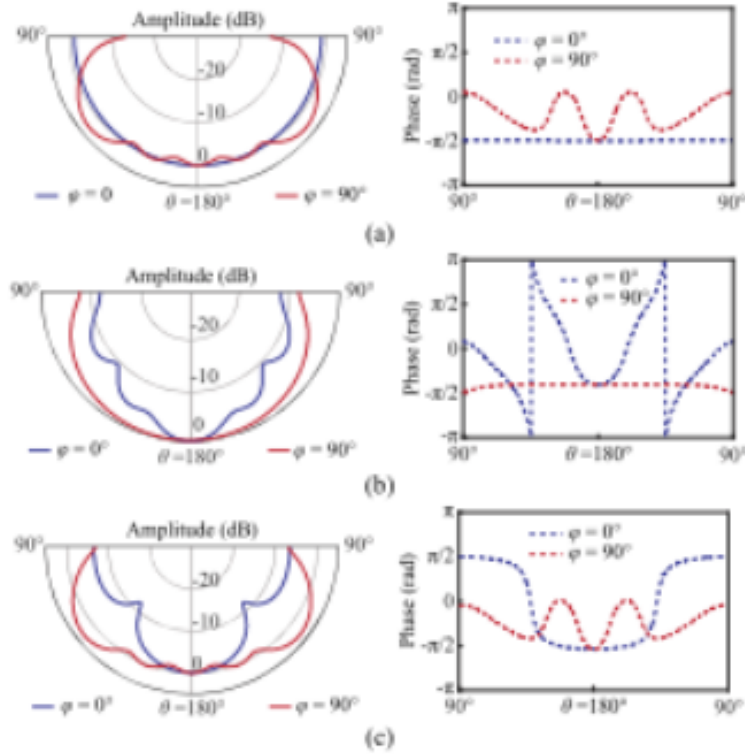


图 5 通过 (6) 式计算的不同孔径尺寸下的散射方向图。(a) $w = 2$ cm, $l = 15$ cm。(b) $w = 15$ cm, $l = 2$ cm。(c) $w = 8$ cm, $l = 15$ cm。

4.4 双基地雷达检测到的多普勒信号

一个单输入多输出 (SIMO) DRS 可被视为多个共享同一发射通道的双基地雷达的组合。图 6 展示了一个用于检测运动物体的双基地雷达。某物体从 (x_S, y_S, z_S) 位置以速度 $\mathbf{v} = (v_x, v_y, v_z)$ 和位移 $\mathbf{p} = (p_x, p_y, p_z)$ 移动。径向速度 $\mathbf{v}_T$ 和 $\mathbf{v}_R$ 是 $\mathbf{v}$ 在矢量 $\mathbf{r}_T$ 和 $\mathbf{r}_R$ 上的投影，因此对应往返距离 $r(t) = |\mathbf{r}_T(t)| + |\mathbf{r}_R(t)|$ 。

在一个短时间间隔 Δt 内，其中 $\Delta \mathbf{p}(t) = (v_x(t), v_y(t), v_z(t))\Delta t$

$$r(t + \Delta t) = \sqrt{|\mathbf{r}_T(t) + \Delta \mathbf{p}(t)|^2} + \sqrt{|\mathbf{r}_R(t) - \Delta \mathbf{p}(t)|^2}. \quad (6)$$

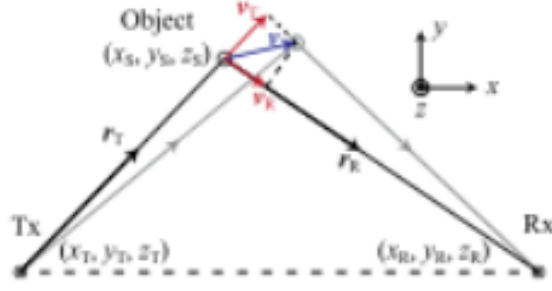


图 6 双基地 DRS 检测到的多普勒相位

然后, $r(t)$ 的导数可计算为 (见附录)

$$\begin{aligned} \frac{dr(t)}{dt} &= \lim_{\Delta t \rightarrow 0} \frac{r(t + \Delta t) - r(t)}{\Delta t} = \mathbf{v}(t) \cdot \hat{\mathbf{r}}_T - \mathbf{v}(t) \cdot \hat{\mathbf{r}}_R \\ &= |v_T(t)| - |v_R(t)|. \end{aligned} \quad (7)$$

根据多普勒公式 [41], 多普勒频率偏移可计算为

$$f_d(t) = (|v_R(t)| - |v_T(t)|)f_c/c \quad (8)$$

其中 f_c 为载波频率, c 为光速。根据 (10) 和 (11)

$$r(t) = -\frac{c}{2\pi f_c} \int 2\pi f_d(t) dt = -\frac{c}{2\pi f_c} \Phi(t). \quad (9)$$

公式 (12) 显示了往返距离 $r(t)$ 与多普勒信号 $-\Phi(t)$ 之间的等效关系。

4.5 由 $\Phi(t)$ 携带的确定性多普勒信号

对于图 2 中布置的天线 T 、 R_1 、 R_2 , 和 R_3 , 我们假设它们的坐标分别为 $(-a, a, 0)$ 、 $(a, a, 0)$ 、 $(a, -a, 0)$ 和 $(-a, -a, 0)$, 一个“向右”手势在 $z = h$ 平面上移动。其轨迹可描述为 $X(x_g, y, h)$, 其中 x_g 是该手势散射相位中心的 x 坐标, y 是该手势恒定的 y 坐标。发射器、移动手势与 TDM 接收器之间的往返距离可表示为

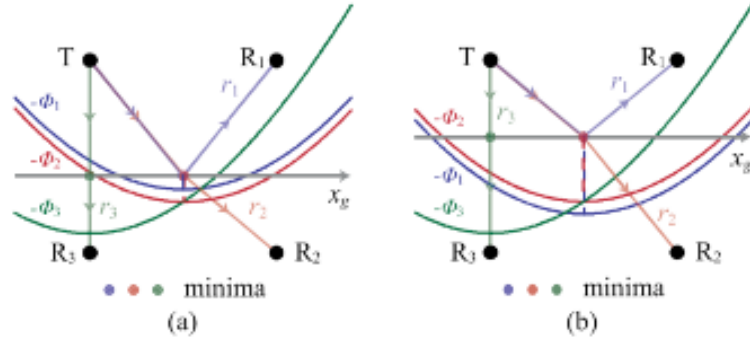


图 7 TDM-SIMO DRS 在“向右”手势中检测到的多普勒信号变化。计算中， $a = 2.5$ cm， $h = 10$ cm， x_g 表示手势从 -4.5 移动到 4.5 cm， y_1 和 y_2 是“向右”手势的 y 坐标，在示例中 (a) $y_1 = -0.5$ cm，(b) $y_2 = 0.5$ cm。

$$\begin{cases} r_1 = \sqrt{(x_g + a)^2 + (y - a)^2 + h^2} \\ \quad + \sqrt{(x_g - a)^2 + (y - a)^2 + h^2} \\ r_2 = \sqrt{(x_g + a)^2 + (y - a)^2 + h^2} \\ \quad + \sqrt{(x_g - a)^2 + (y + a)^2 + h^2} \\ r_3 = \sqrt{(x_g + a)^2 + (y - a)^2 + h^2} \\ \quad + \sqrt{(x_g + a)^2 + (y + a)^2 + h^2}. \end{cases} \quad (10)$$

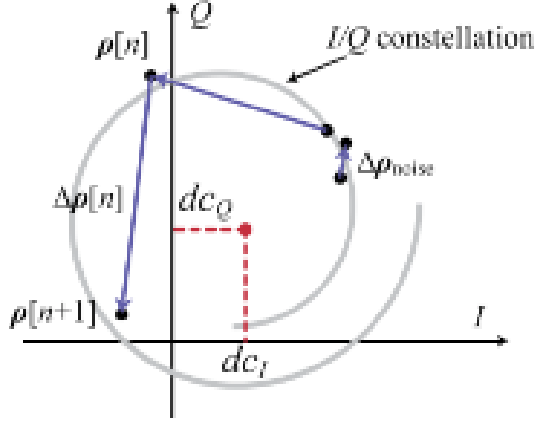
根据 (12) 和 (13)，可以计算出相应的多普勒信号 $-\Phi_i(t)$ ，如图 7 所示，其中 $i = 1, 2, 3$ 。可以看出，对于“向右”手势， $-\Phi_3(t)$ 的最小值首先出现，随后是 $-\Phi_1(t)$ 和 $-\Phi_2(t)$ 的最小值。对于位于正方形角落的天线， $-\Phi_1(t)$ 和 $-\Phi_2(t)$ 的最小值将同时出现。这种确定但不同的多普勒信号也出现在其他手势中。在本研究中，将利用 $-\Phi_i(t)$ 的这些最小值的时间顺序来进行鲁棒的手势识别。

在 (13) 中， h 也表示手势的高度。当 h 发生变化时， $-\Phi_i(t)$ 曲线的形状也会改变。然而，极小值的时间顺序将保持不变。这意味着基于此类极小值时间顺序的手势识别将与手势的高度无关。

4.6 检测信号的分割

为了识别预定义的手势，应首先分离包含所有运动信息的信号，即 $\Phi[n]$ 。这可以通过对检测到的 I/Q 信号进行动态功率分析来实现。

图 8 说明了采样时刻 I/Q 信号在 n 和 $n + 1$ 处的情况。相邻采样期间 I/Q 信号的变化向量 $\Delta \rho[n]$ 表示。当存在移动散射体时，


 图 8 $\rho[n]$ 和 ρ_{noise} 在 I/Q 星座图中的示意图。

$$\Delta \rho[n] = \begin{pmatrix} A_I[n+1] \cos(\Phi[n+1]) - A_I[n] \cos(\Phi[n]) + \Delta n_I[n] \\ A_Q[n+1] \sin(\Phi[n+1]) - A_Q[n] \sin(\Phi[n]) + \Delta n_Q[n] \end{pmatrix}^T \quad (11)$$

反映“ I/Q ”信号的“差分功率” $\Delta P[n]$ I/Q 信号的 Δ ，其中 $\Delta n_I[n]$ 为 $n_I[n+1] - n_I[n]$ ，因此 $\Delta n_Q[n]$ 也是如此。这里， $\Delta P[n]$ 可计算为

$$\Delta P[n] = |\Delta \rho[n]|^2. \quad (12)$$

在没有任何运动的情况下， $\Delta \rho_{noise}[n] = (\Delta n_I[n], \Delta n_Q[n])$ 表示电路噪声的“差分功率”。显然， $\Delta \rho$ 始终大于 $\Delta \rho_{noise}$ ，因此 ΔP 可有效地用于将包含手势信息的信号段与检测到的信号分离。

为确定分割的阈值，可基于 $\Delta \rho_{noise}$ 的数学期望即 $E(\Delta \rho_{noise})$ 来确定，据此可选择略大于该期望的值作为阈值。在本研究中，该值为 $1.1 E(\Delta \rho_{noise})$ 。

为了确保分离的可靠性，可以在每个时间间隔对 $\Delta P[n]$ 执行 m 点移动平均窗口处理，其中

$$\overline{\Delta P_m}[n] \triangleq m^{-1} \sum_{j=n-m/2+1}^{n+m/2} \Delta P[j] \quad (13)$$

以减少由于意外故障导致的潜在误判。如果 $\overline{\Delta P_m}[n]$ 大于阈值，则认为在此时间间隔内存在运动。所有此类连续的时间间隔对应一个信号段，在该信号段期间被认为存在运动。对所有接收通道检测到的信号均可执行相同的操作。所获得三个信号段反映了从不同方向检测到的同一运动。这些信号段的并集将产生一个最宽的信号段，包含完整运动的信息，从中可以识别出潜在的手势。

4.7 TDM 采样的数据对齐

如图 1 所示, $TDM-SIMO$ 架构显著降低了 DRS 的复杂性和成本。然而, 采样得到的数据序列需要进行对齐。

对于 M 个 TDM 接收通道, 检测到的串行 I 通道序列 $I_1[n], I_2[n+1/M], \dots, I_M[n+(M-1)/M], I_1[n+1], I_2[n+1+1/M], \dots$ 应重新对齐为 $I_{1,n}, I_{2,n}, \dots, I_{M,n}, I_{1,n+1}, I_{2,n+1}, \dots$ 。第 i 个通道中, 检测信号与对齐信号之间的差异为 $(i-1)T + 1/M$, 其中 T_{TDM} 是相邻 TDM 采样之间的时间间隔。在本研究中, 设计了一组数字相移滤波器 TDM 来补偿这种时间不同步, 从而可得到第 i 个通道该滤波器的 z 域响应为 [42]。其频率响应为

$$H_i(e^{j\omega}) = H_i(z)|_{z=e^{j\omega}} = e^{j(i-1)\omega/M} \quad (14)$$

这种全通相位偏移滤波器可以被综合为

$$\begin{aligned} h_i(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j(i-1)\omega/M} e^{j\omega n} d\omega \\ &= \frac{\sin\{\pi[n + (i-1)/M]\}}{\pi[n + (i-1)/M]}. \end{aligned} \quad (15)$$

将 $I_i[n]$ 与 $h_i(n)$ 进行卷积后, 来自不同通道的采样数据可以对齐。类似地, 检测到的 Q 通道信号也可以对齐。

4.8 分段收拾的重缩放与拟合

在完成分割和数据对齐之后, 可以获得连续的时间段, 每段包含一个单一的手势。对于最宽的手势段, 其实际持续时间或总采样数因手势而异。此外, 检测到的原始多普勒相位信号 Φ_i 可能是锯齿状的, 因此需要对手势分段信号进行重缩放和拟合, 以获得长度一致且形状平滑的规范化手势信号, 便于后续识别。

在本研究中, 所有最宽的段均被重缩放至 $(0, 1)$ 区间。为了保持 $-\Phi_i$ 的形状并防止毛刺的影响, 本文采用八阶多项式拟合, 从而使信号 $-\Phi_i$ 上的噪声不会被拟合。

在重缩放和拟合之后, 第 i 个通道的 $-\Phi_i$ 极小值时刻 (如 II-E 节所述) 可在区间 $(0, 1)$ 内量化, 并表示为 $t_{m,i}$ 。在本研究中, 极小值时刻以向量形式表示为 $\mathbf{t}_m = (t_{m,1}, t_{m,2}, t_{m,3})$, 可用于表征每种手势, 并将作为后续讨论的分类器的输入。

经过此次重缩放后, 极小值时刻与手势速度无关。这在识别快速和慢速手势

方面起着重要作用。

4.9 基于分类的手势识别

分类是机器学习中常用的技术之一 [23]，旨在将一组数据进行分类到各个类别中。它已应用于图像识别、语音处理、生物信息学、自然语言处理等领域 [24]。

迄今为止，已有多种分类算法或分类器被提出。主流分类器包括逻辑回归、线性判别分析 (*LDA*)、决策树、朴素贝叶斯分类器和支持向量机 (*SVM*) [43]。

在本研究中，选择用于多分类的一对多 (*OVR*) *SVM* 来识别预定义的多类手势。信号处理、基于已知手势数据集训练 *OVR SVM* 以及对未知手势进行识别的流程图分别如图 9(a)-(c) 所示。

为了训练 *OVR SVM*，可以通过对每个已定义的手势检测 N 次来准备包含已知手势的数据集。然后，该数据集可复制成 C 份，其中 C 为已定义手势的数量。在第 i 个 *SVM* 的训练中，所有第 i 个手势被标记为 $+1$ ，其余所有手势均被标记为 -1 。

基于训练好的 *SVM*，可通过比较每个 *SVM* 评估出的“得分”或相似度来识别新手势。如果第 i 个 *SVM* 给出的得分最高，则该手势将被识别为第 i 个手势。更多细节见 [23]。

5 仿真

为了验证第三节中提出方法的有效性，我们根据图 9 所示的步骤，基于推导出的方程进行了数值仿真。在所有仿真中，发射天线和接收天线 T 、 R_1 、 R_2 和 R_3 的坐标分别为 $(-2.5, 2.5, 0)$ 、 $(2.5, 2.5, 0)$ 、 $(2.5, -2.5, 0)$ 和 $(-2.5, -2.5, 0)$ ；简化手掌模型的 L 、 W 和 T 分别为 15、8 和 2，单位均为厘米。

5.1 不同手掌方向检测到的多普勒信号

定义的手势可分为两种类型。对于“左”、“右”、“前”、“后”和“对角”手势，手掌垂直于天线平面，以较窄的孔径散射照射场。对于“轻触”手势，手掌平行于天线平面，以大得多的孔径进行散射。

为解释定义这两类手势的原因，我们比较了手掌垂直和平行时执行“右”手势所计算出的多普勒信号。在计算中，两个散射孔径中心即 (X_s, y, h) 分别从 $(-20, 0, 10)$ 移动到 $(20, 0, 10)$ 。根据公式 (2)-(8)， I/Q 信号的星座图、差分功率 $\Delta P[n]$ ，以及恢复的多普勒相位 $-\Phi[n]$ ，如图 10 所示。

可以看出，垂直和平行移动的手掌会产生非常不同的 I/Q 信号和多普勒相

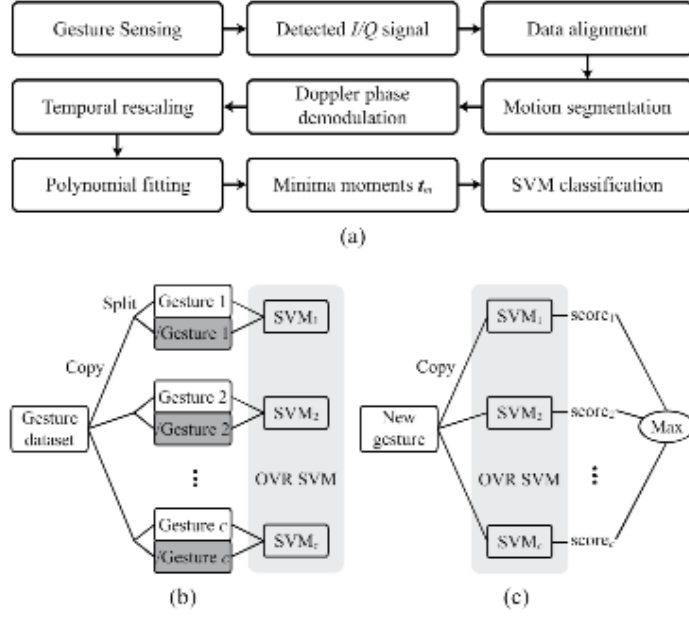


图9 手势识别流程图。(a) 信号处理。(b) OVR SVM 的训练。(c) 基于已训练 OVR SVM 的手势识别。

位。特别是对于平行移动的手掌，图 10(c) 中的极小值几乎无法确定，而图 10(f) 中的极小值则很容易识别。这正是定义在天线平面上方平行移动手势时所考虑的因素。

5.2 按照流程图进行仿真

在本节中，将按照图 9(a) 所示步骤进行数值仿真。

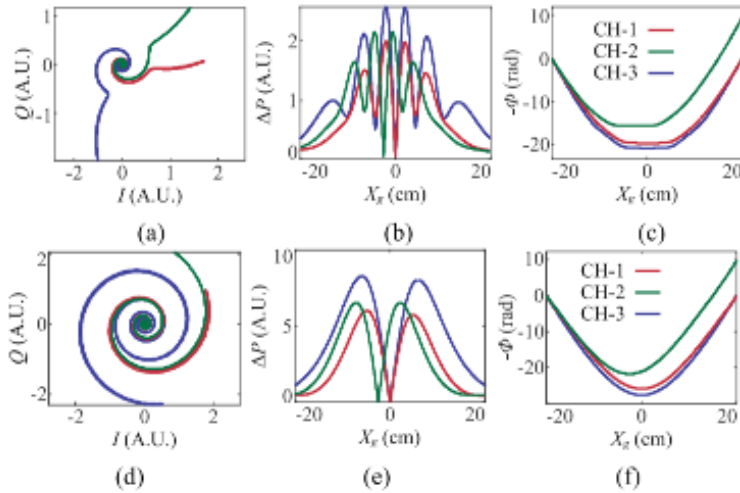


图10 针对掌心不同朝向模拟生成的多普勒信号。(a) - (f) 分别为 I/Q 信号的星座图、分段后的差分功率以及平行和垂直掌心情况下恢复的多普勒相位。

首先，使用与计算图 10 相同的方法，也可以计算出由 R_1 , R_2 , 和 R_3 天线分别“检测”到的两个连续的“向右”和“向左”手势所产生的 I/Q 信号。结果如

图 11 所示。不同之处在于添加了白高斯噪声，使得 I/Q 信号能够在 20 dB 信噪比 (SNR) 下生成。

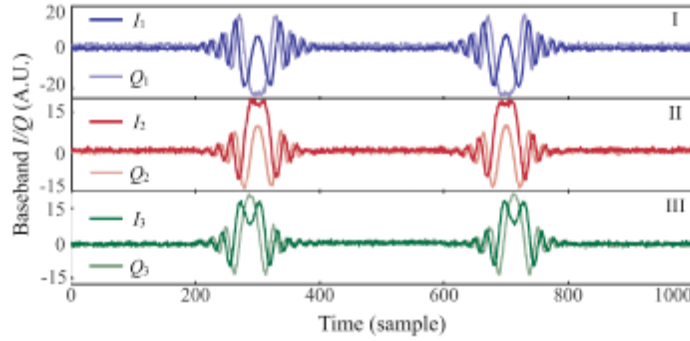


图 11 由 R1, R2, 和 R3 天线分别“检测”到的两个连续“向右”和“向左”手势所产生的模拟 I/Q 信号。

然后，根据第三节-F 对这些信号进行分段。图 12 中的面板 I 和 II 分别显示了解调后的多普勒相位以及 60 点移动平均微分功率 $\Delta P_{60}[n]$ 。选择阈值为 2 时，手势可以在灰色标记的区域中被分割。可以看出，两个手势都被正确地分割。面板 III 展示了在更差的 10 dB 信噪比条件下与面板 II 相同的计算结果。可以看出，手势片段几乎无法识别。

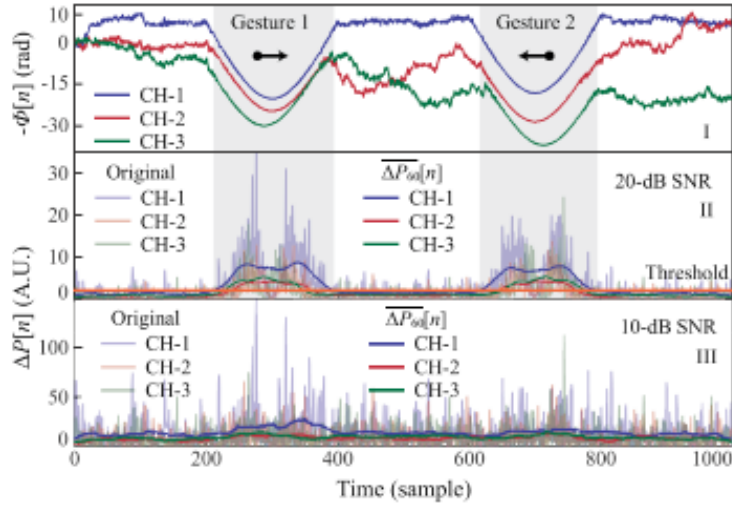


图 12 两个连续“向右”和“向左”手势的分段结果。子图 I：20 dB SNR 下的解调多普勒相位。子图 II：20 dB SNR 下模拟基带信号的 $P[n]$ 和 $P_{60}[n]$ ，以及用于分段的阈值。子图 III：10 dB SNR 下模拟基带信号的 $P[n]$ 和 $P_{60}[n]$ 。

接下来，基于第三节-H 的讨论，通过八阶多项式拟合对已分割的手势进行重缩放。如图 13 所示，在重缩放和拟合之后，可以清晰提取重缩放后多普勒相位的极小值时刻，并据此确定这些极小值的时间顺序。在本研究中，这种在通道顺序上确定且量化的极小值，例如该图中的 $t_m = (0.48, 0.48, 0.42)$ 和 $t_m =$

(0.47, 0.47, 0.54)，被用作 *OVR SVM* 分类器的输入，而不是“黑箱”手势分类中使用的 *I/Q* 星座图或多普勒特征。

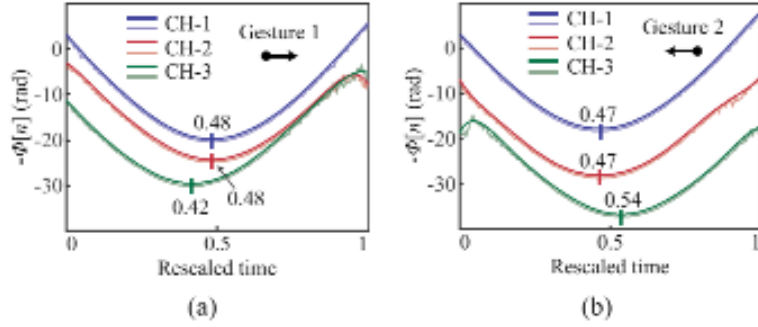


图 13 连续“右”和“左”手势的缩放并拟合后的多普勒相位中多普勒极小值的确定。深色线条：八阶多项式拟合的多普勒相位。浅色线条：含噪声的原始多普勒相位。(a) 手势 1 的量化极小值时刻为 $t_m = (0.48, 0.48, 0.42)$ ，(b) 手势 2 的量化极小值时刻为 $t_m = (0.47, 0.47, 0.54)$ 。

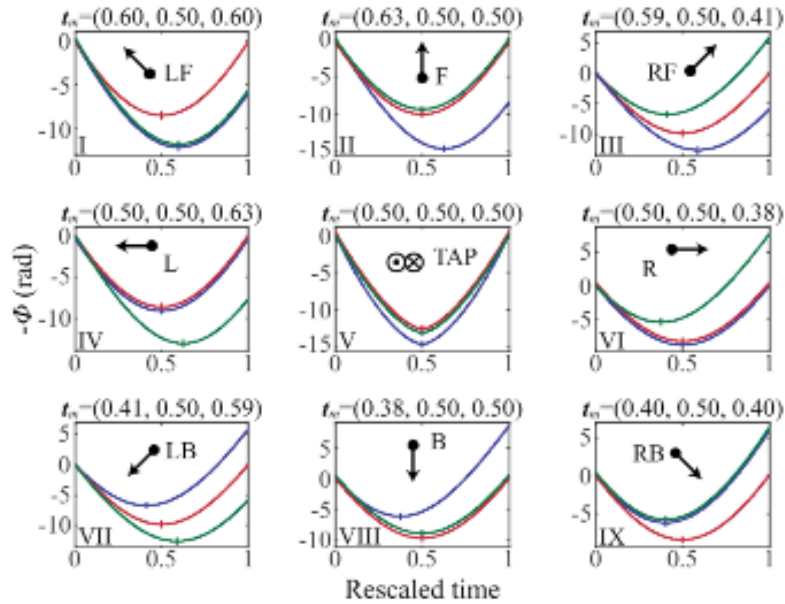


图 14 不同接收通道检测到的定义手势的模拟多普勒相位。

图 14 显示了从为所有定义的手势模拟的 *I/Q* 信号中分割出的多普勒相位。在仿真中，对于所有水平方向移动的手势，手底边与天线平面之间的距离设为 10 cm；对于敲击手势，手掌与天线平面之间的最短距离设为 5 cm。可以看出，每个分割手势的极小值的时间序列顺序是确定的。

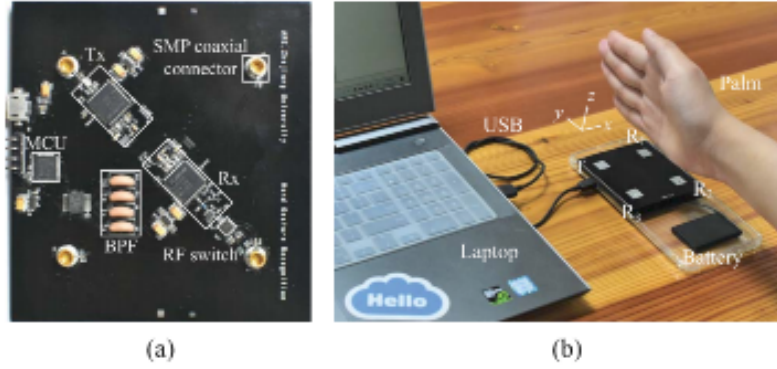


图 15 使用一对 Wi-Fi 芯片实现的时分双工单输入多输出 DRS。(a) 射频板。(b) 电池供电的 DRS 及实验环境。

6 实验

6.1 时分双工单输入多输出 DRS 的实现

图 15 展示了根据图 1 提出架构实现的 *TDM-SIMO DRS*。在本研究中，使用了一对 *Wi-Fi* 芯片 *Maxim Integrated MAX2829* 来模拟 5.8 GHz 的 *LTE-U* 和 *Wi-Fi* 信道，并使用射频开关 *HMC7992* 将接收信道分别切换至天线 R_1 , R_2 , 和 R_3 。类似于 [44]，采用 *L* 形排列的接收天线用于检测三维手势。在不失一般性的前提下，这些天线被设计为标准贴片天线。天线板与射频板通过四个超小型推入式连接器 (*SMP*) 同轴连接器相连。一个标准锂电池为 *DRS* 供电。采用 *STM32F103T8U6* 微控制器单元 (*MCU*) 对 *Wi-Fi* 芯片进行配置，并通过 *MCU* 内置的 *ADC* 采集 *I/Q* 信号。采样数据最终通过安装在射频板上的 *Silicon Labs CP2102* 通用串行总线 (*USB*) -通用异步收发传输器 (*UART*) 桥接芯片传输到笔记本电脑。

6.2 实验手势信号的处理

对应于图 11-13，图 16 中的面板 I-III 分别显示了天线 R_1 , R_2 , 和 R_3 检测到的基带 *I/Q* 信号，依次对应连续的“前”、“后”、“左”和“右”手势。采用相同的处理方法计算得到图 12 和图 17，展示了图 16 中检测到的手势分割结果。图 18 显示了经解调后的多普勒相位及其极小值时刻 t_m ，以便进一步分类。

可以看出，对于实验数据，以“向右”手势为例，由于实际手部的非平面散射孔径， $-\Phi_1(t)$ 和 $-\Phi_2(t)$ 的最小值可能出现的时间略有不同。然而，如果用于训练分类器的数据集包含了此类偏移的最小值，则其对分类的准确性影响较小。

对应于图 14，相同九种手势在距离天线平面相同距离处执行，*TDM-SIMO DRS*

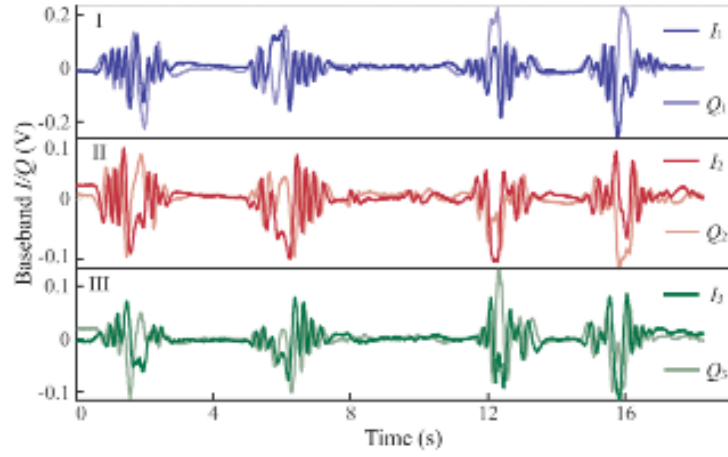


图 16 实验 I/Q 信号来自天线 R1、R2 和 R3，分别检测到的连续四次“向前”、“向后”、“向左”和“向右”手势。

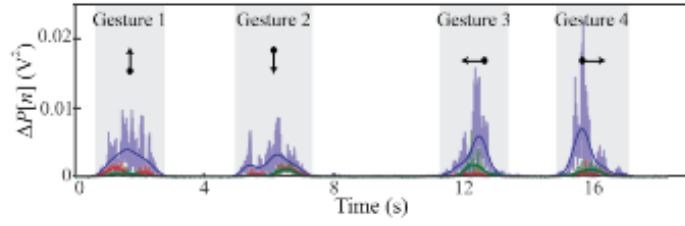


图 17 基于图 16 中检测到的信号的微分功率 $P[n]$ 以及移动平均 $P_{60}[n]$ 进行的分割过程。

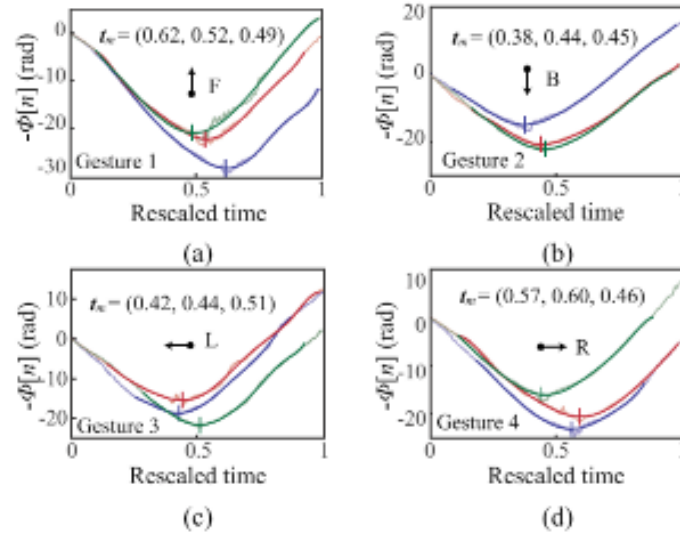


图 18 图 16 中连续四个手势的重缩放并拟合后的多普勒相位多普勒极小值的确定。深色线条：八阶多项式拟合的多普勒相位曲线。浅色线条：原始多普勒相位曲线。四个手势的量化极小值时刻分别为 (0.62, 0.52, 0.49)、(0.38, 0.44, 0.45)、(0.42, 0.44, 0.51) 和 (0.57, 0.60, 0.46)。

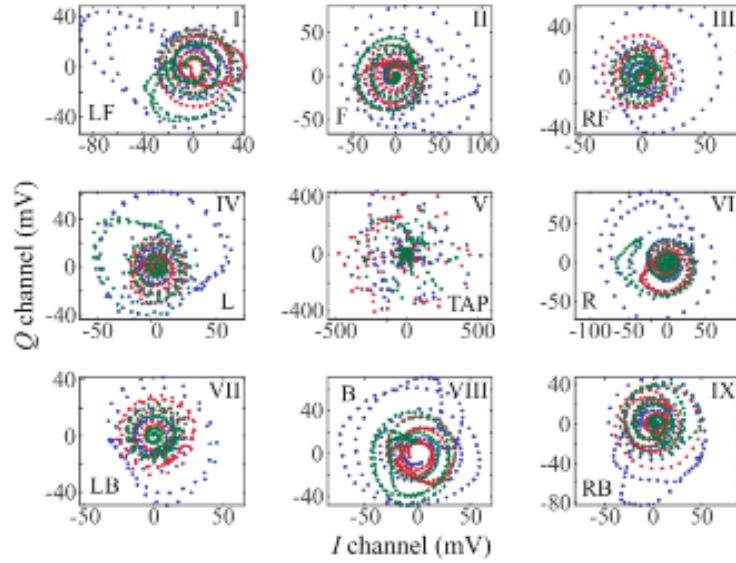


图 19 实验中执行手势的星座图。

检测到的这些手势对应的 I/Q 信号星座图分别如图 19 所示。可以注意到，由于手掌的垂直运动和更大的散射面积，“点击”手势的幅度和形状与其他手势明显不同。图 20 显示了从图 19 绘制的星座图中检索出的 $-\Phi_i$ 。可以看出，极小值点

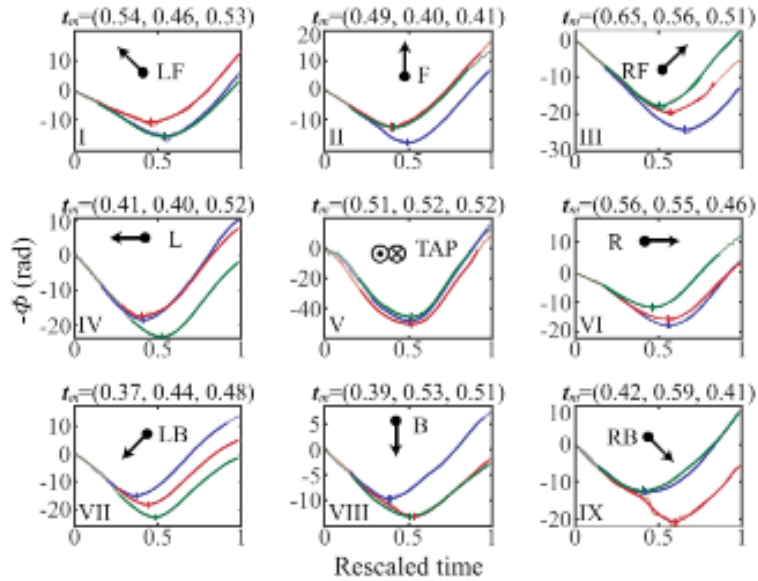


图 20 执行手势的实验多普勒相位变化及极小值时刻。

的顺序 t_m 与图 14 中的顺序一致。

6.3 手势分类

为了对定义的手势进行分类，我们为不同的手势收集了 500 个样本作为数据集，每个手势 50 到 60 个样本。其中，500 个样本中的 80%（每个手势约 40 到 50 个样本）被随机选作训练集。剩余的 20% 被选作测试集，包括待识别的未

知手势。为了可视化训练集和测试集中的极小值时刻 t_m ，建立了一个矩形坐标系，其坐标轴对应于每个接收通道检测到的极小值时刻。如图 21 所示，在两个数据集中，每种手势的极小值时刻均被聚类，并且与其他聚类明显分离，验证了由极小值时刻表示的手势具有确定性。这也意味着，使用训练集训练得到的 *OVR SVM* 可以准确分类测试集中的手势。

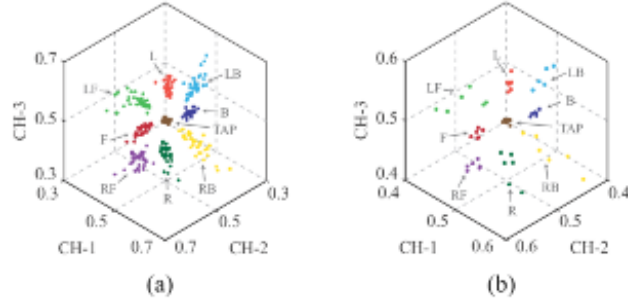


图 21 将各类手势的多普勒相位极小值时刻 t_m 映射到三维特征坐标系中。坐标轴表示通道 1-3 的重缩放时间。(a) 训练集。(b) 测试集。

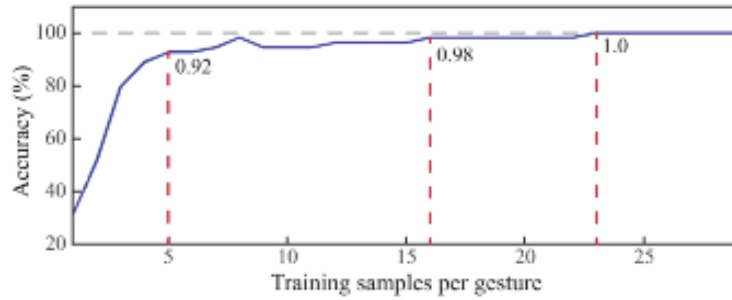


图 22 通过未知样本评估的 *OVR SVM* 学习曲线。

为了评估基于极小值时刻分类的鲁棒性，我们通过使用不同子集训练得到的 *OVR SVM* 对 50 个未知手势进行分类，计算出学习曲线 [23]。结果如图 22 所示。可以看出，仅用六个样本训练的 *OVR SVM* 即可达到 98% 的识别准确率，当训练样本增加到 23 个时，可获得 100% 的准确率。使用普通个人计算机进行训练耗时不到 3 s。

为了验证该分类器的泛化能力，通过邀请另外四名受试者执行相同的 *gesture* 进行了一次留一受试者实验 [12]。每个手势重复 10 次。在执行之前，他们了解每个手势的移动方向、手部姿势以及敏感区域。图 23 显示了由这些受试者检测到的极小值时刻组成的数据集的映射情况。可以看出，与图 21(a) 所示训练集的映射相比，这些极小值时刻也呈现出聚类且明显分离的状态。这表明，尽管不同受试者执行的手势之间存在差异，但经过重新缩放的极小值时刻对于 *OVR SVM* 分类器而言仍表现出足够的相似性。

使用基于原始受试者检测数据集训练的分类器，对图 23 中其他受试者检测到的最小动作时刻进行分类。通过计算每个新受试者的混淆矩阵，可分别获得 96.67%、100%、97.78% 和 96.67% 的识别准确率，该结果验证了由某一特定受试者训练得到的分类器适用于未知受试者。

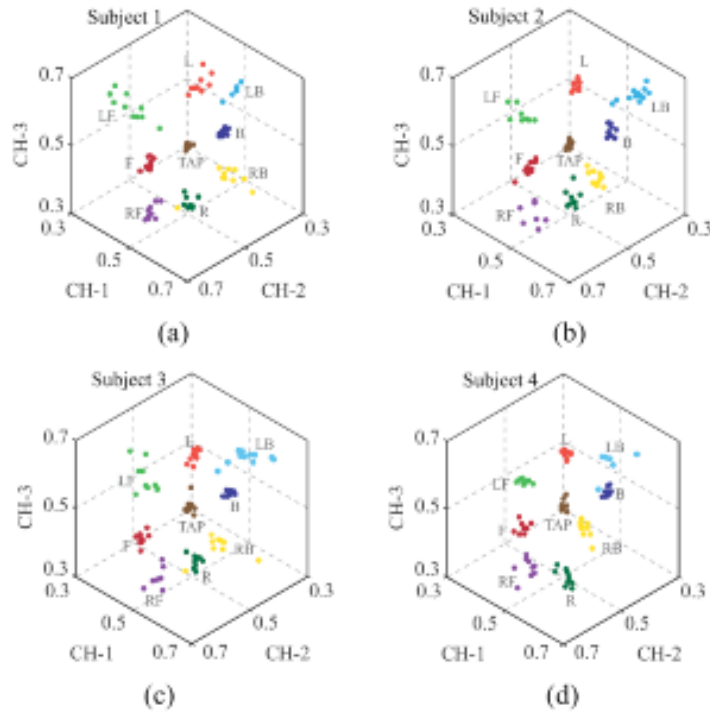


图 23 从新受试者检测到的最小值时刻的映射。

最后进行了交叉验证 [23] 实验，其中将来白全部五名受试者的检测手势混合在一起。在混合的手势中，随机选择 75% 的手势作为训练集，其余 25% 作为测试集。对于每种手势，测试集中的手势数量为 24 个。图 24 显示了针对参与单个受试者留出实验的五名受试者所执行手势计算得到的混淆矩阵。可计算得出，总体识别准确率高达 98.6%。此外，图 24(b) 展示了 100 次试验中交叉验证准确率的直方图。经计算，最大、最小和平均准确率分别为 99.54%、94.44% 和 97.83%，标准差为 1.15%。

6.4 讨论

上述结果验证了所提出的基于检测和处理确定性多通道多普勒信号的智能设备方法的有效性和准确性。

作为消费产品，智能手机等智能设备高度集成、紧凑，并且对电池寿命、产品尺寸和成本极为关注。该连续波解决方案工作在单一频率，可通过重用智能设备现有的射频资源来实现，增加的成本、功耗和空间占用均可忽略不计。

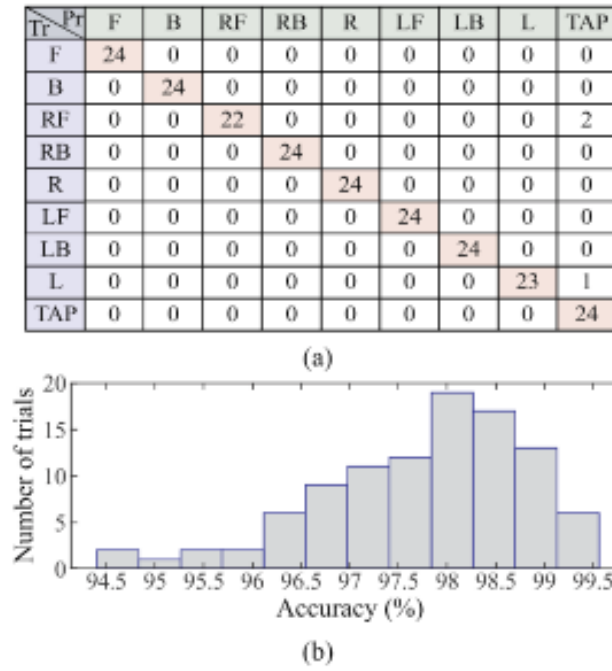


图 24 训练后的 OVR SVM 准确性评估。(a) 混淆矩阵, 其中 Tr 和 Pr 分别代表“真实”和“预测”。(b) 100 次试验中交叉验证准确率的直方图。

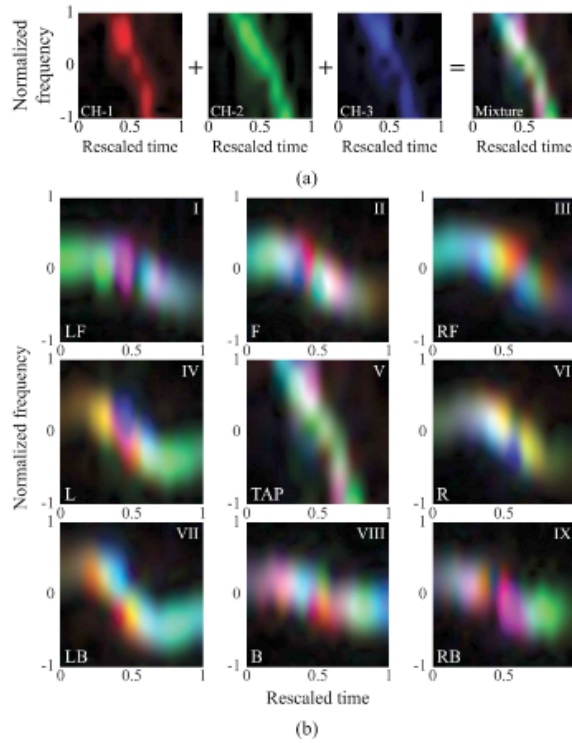
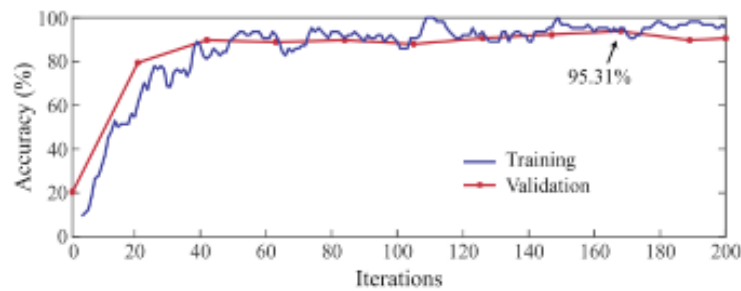


图 25 不同手势产生的时频谱图。(a) 单色和全色谱图。(b) 用于基于 DCNN 分类的全色多普勒谱图。

多普勒信号的不确定性带来了一个独特特征，即使用极小数据集训练的高效 *OVR SVM* 分类器即可实现高识别准确率。相比之下，基于对多普勒信号时频谱进行分类的手势识别则需要多得多的训练样本。

作为对比，图 25(a) 展示了 *SIMODRS* 每个通道检测到的多普勒相位的短时傅里叶变换 (*STFT*)。在计算该 *STFT* 谱图时，对每个通道使用了 20 点的汉宁滑动窗，重叠点数为 10。由于每个手势的时间持续长度不同，因此将谱图调整为 400×400 像素。然后，将每个通道的谱图强度归一化到 (0,1) 范围内。

将获得的时频谱绘制成单色红、绿、蓝三色图像，然后混合生成全色 *spectrogram*。图 25(b) 展示了基于图 19 所示信号计算出的九种不同手势的 *spectrogram*。在以往基于多普勒雷达的识别 [10], [12], [14], [16], [17], [25] 中，此类“图像”被用于训练分类器，通常为深度卷积神经网络 (*DCNN*)，之后可通过训练好的 *DCNN* 识别未知手势。



(a)

Tr \ Pr	F	B	RF	RB	R	LF	LB	L	TAP
F	24	0	1	0	0	0	0	0	0
B	0	25	0	0	0	0	1	0	0
RF	0	0	21	0	1	0	0	0	0
RB	0	0	2	12	2	1	0	0	0
R	0	0	0	0	22	0	0	0	0
LF	0	0	0	0	0	21	0	0	0
LB	0	0	0	0	0	0	19	0	0
L	0	0	0	0	0	0	1	19	0
TAP	0	0	0	0	0	0	0	0	20

(b)

图 26 与基于 *DCNN* 分类的比较。(a) *DCNN* 分类器的训练。(b) 混淆矩阵。

例如，使用检测到的手势的 560 个频谱图来训练适用于移动和嵌入式设备的“*MobileNets*” *DCNN* [45]，并采用大小为 8 的小批量以避免潜在的过拟合。这些频谱图来自全部五名受试者的混合样本。然后，通过训练好的 *DCNN* 对 192 个未知频谱图进行分类，从而获得识别准确率随迭代次数的变化情况。

如图 26(a) 所示，随着迭代次数的增加，识别准确率总体呈上升趋势。最终

的识别准确率达到 95.31%。在普通计算机上, 基于频谱图训练 *DCNN* 耗时约 8 分钟, 远长于基于序列极小值矩训练 *OVR SVM* 所用的 3 秒。尽管 *DCNN* 需要更大的训练集, 但其难以达到 100% 的识别准确率。基于训练好的 *DCNN*, 可计算出如图 26(b) 所示的混淆矩阵。

实际手势识别的另一个挑战是处理干扰性的随机手势和动作。在本研究中, 尽管 *OVR SVM* 能够帮助过滤掉其中一部分, 但该问题主要在信号分割阶段得以解决。如图 20 所示, 对于每一个符合条件的手势, 由于每个接收天线检测到的多普勒相位分割曲线是凸函数, 因此仅存在一个最小值时刻。因此, 只需检查分割信号中最小值时刻的数量, 即可排除随机手势或动作。

最后, 本研究所定义的手势对应于多点触控屏幕的主要功能, 目的是为智能设备提供额外的无线手势控制功能。这些手势的多普勒相位曲线均为凸函数。我们的方法也可扩展至双向和多向手势, 其多普勒相位曲线由多个相连的凸函数组成。这类手势可以是在水平面或垂直面内执行的手势组合。然而, 现有方法不适用于无法提取出有序最小值时刻的手势。对于诸如手指摩擦等手势, 基于微多普勒的分析可能是必不可少的。

7 结论

总之, 我们从理论上分析并实验验证了基于单输入多输出差分雷达系统 (*SIMO-DRS*) 的手势识别方法, 该方法可通过轻微修改智能设备现有的射频前端来实现。通过检测和处理由 *SIMO* 雷达系统探测到的确定性多普勒信号, 仅需少量数据集即可训练一个 *OVR SVM*, 并实现九种预定义手势的毫秒级识别。我们设想这种鲁棒且高效的方法还可扩展到多种应用中。

8 附录

8.1 公式 (7) 的推导

对于一个 y 极化的赫兹偶极子, 记其矢量势为

$$\mathbf{A} = \hat{\mathbf{y}} \mu I \Delta l e^{-jkr} / 4\pi r \quad (1)$$

其中 Δl 和 I 分别为无穷小长度和线电流强度。在 r - θ - φ 球坐标系中

$$\hat{\mathbf{y}} = \hat{\mathbf{r}} \sin \theta \sin \varphi + \hat{\boldsymbol{\theta}} \cos \theta \sin \varphi + \hat{\boldsymbol{\varphi}} \cos \varphi \quad (2)$$

然后，电场

$$\begin{aligned}
 \mathbf{E} &= \frac{1}{j\omega\epsilon} \nabla \times \mathbf{H} = \frac{1}{j\omega\epsilon} \nabla \times \left(\frac{1}{\mu} \nabla \times \mathbf{A} \right) \\
 &= \frac{1}{j\omega\epsilon} \frac{I\Delta l}{4\pi} \left[\frac{2}{r^3} (1 + jkr) \sin \theta \sin \varphi e^{-jkr} \hat{\mathbf{r}} \right. \\
 &\quad + \left(\frac{k^2}{r} - \frac{1}{r^3} - \frac{jk}{r^2} \right) \cos \theta \sin \varphi e^{-jkr} \hat{\boldsymbol{\theta}} \\
 &\quad \left. + \left(\frac{k^2}{r} - \frac{1}{r^3} - \frac{jk}{r^2} \right) \cos \varphi e^{-jkr} \hat{\boldsymbol{\phi}} \right]
 \end{aligned} \tag{3}$$

在远场区域

$$\begin{aligned}
 \lim_{r \rightarrow \infty} \mathbf{E} &= \frac{1}{j\omega\epsilon} \frac{I\Delta l}{4\pi} \frac{k^2}{r} e^{-jkr} (\hat{\boldsymbol{\theta}} \cos \theta \sin \varphi + \hat{\boldsymbol{\phi}} \cos \varphi) \\
 &= \frac{-jk\eta_0 I\Delta l e^{-jkr}}{4\pi r} (\hat{\boldsymbol{\theta}} \cos \theta \sin \varphi + \hat{\boldsymbol{\phi}} \cos \varphi)
 \end{aligned} \tag{4}$$

在直角坐标系中

$$\begin{aligned}
 \hat{\boldsymbol{\theta}} &= \hat{\mathbf{x}} \cos \theta \cos \varphi + \hat{\mathbf{y}} \cos \theta \sin \varphi - \hat{\mathbf{z}} \sin \theta \\
 \hat{\boldsymbol{\phi}} &= -\hat{\mathbf{x}} \sin \varphi + \hat{\mathbf{y}} \cos \varphi
 \end{aligned} \tag{5}$$

那么，赫兹偶极子散射的电场的 y 分量为

$$E_{dipole}^y = \frac{-jk\eta I\Delta l e^{-jkr}}{4\pi r} (\cos^2 \theta \sin^2 \varphi + \cos^2 \varphi) \tag{6}$$

将 $I\Delta l$ 替换为实际的表面电流元 $J(\zeta, \xi)d\zeta d\xi$ ，该赫兹偶极子辐射的远场电场可推导为

$$\begin{aligned}
 dE_S^y(\zeta, \xi, r, \theta, \varphi) &= \frac{-jk\eta_0 J(\zeta, \xi) e^{-jkr}}{4\pi r} (\cos^2 \theta \sin^2 \varphi + \cos^2 \varphi) d\zeta d\xi.
 \end{aligned} \tag{7}$$

8.2 公式 (9) 和 (10) 的推导

对于图 6 所述的双基地雷达，瞬态位移 $\mathbf{p} = (p_x, p_y, p_z)$ 和三维瞬态速度 $\mathbf{v} = (v_x, v_y, v_z)$ 满足 $\mathbf{v} = d\mathbf{p}/dt$ ，其微分形式是 $\Delta\mathbf{p}(t) = \mathbf{v}(t)\Delta t$ 。因此，往返距离在时间 t 和 $t + \Delta t$ 时可分别计算为 $r(t) = |\mathbf{r}_T(t)| + |\mathbf{r}_R(t)|$ 和

$$r(t + \Delta t) = \sqrt{|\mathbf{r}_T(t) + \Delta\mathbf{p}(t)|^2} + \sqrt{|\mathbf{r}_R(t) - \Delta\mathbf{p}(t)|^2} \tag{8}$$

当 Δt 趋近于 0 时

$$\begin{aligned}
& \lim_{\Delta t \rightarrow 0} r(t + \Delta t) \\
&= \lim_{\Delta t \rightarrow 0} \sqrt{|\mathbf{r}_T(t)|^2 + 2\mathbf{r}_T(t) \cdot \Delta \mathbf{p}(t) + |\Delta \mathbf{p}(t)|^2} \\
&\quad + \sqrt{|\mathbf{r}_R(t)|^2 - 2\mathbf{r}_R(t) \cdot \Delta \mathbf{p}(t) + |\Delta \mathbf{p}(t)|^2} \\
&\approx \lim_{\Delta t \rightarrow 0} \sqrt{|\mathbf{r}_T(t)|^2 + 2\mathbf{r}_T(t) \cdot \Delta \mathbf{p}(t)} \\
&\quad + \sqrt{|\mathbf{r}_R(t)|^2 - 2\mathbf{r}_R(t) \cdot \Delta \mathbf{p}(t)} \\
&\approx \lim_{\Delta t \rightarrow 0} |\mathbf{r}_T(t)| + \Delta \mathbf{p}(t) \cdot \hat{\mathbf{r}}_T(t) + |\mathbf{r}_R(t)| - \Delta \mathbf{p}(t) \cdot \hat{\mathbf{r}}_R(t)
\end{aligned} \tag{9}$$

注意，在第一次近似中， $|\Delta \mathbf{p}(t)|^2$ 已被忽略。在第二次近似中，应用了二阶泰勒展开式 $\lim_{\alpha \rightarrow 0} (1 + \alpha)^{1/2} \approx 1 + \alpha/2$ 。因此，往返距离的导数可推导为

$$\begin{aligned}
\frac{dr(t)}{dt} &= \lim_{\Delta t \rightarrow 0} \frac{r(t + \Delta t) - r(t)}{\Delta t} \\
&= (\Delta \mathbf{p}(t) \cdot \hat{\mathbf{r}}_T - \Delta \mathbf{p}(t) \cdot \hat{\mathbf{r}}_R) / \Delta t \\
&= \mathbf{v}(t) \cdot \hat{\mathbf{r}}_T - \mathbf{v}(t) \cdot \hat{\mathbf{r}}_R \\
&= |v_T(t)| - |v_R(t)|
\end{aligned} \tag{10}$$

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四、外文原文

Hand Gesture Recognition for Smart Devices by Classifying Deterministic Doppler Signals

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Abstract—Personal devices such as smartphones and tablets are rapidly becoming personal communication, information, and control centers. Apart from multitouch screens, human gestures are considered as a new interactive human–smart device interface. In this work, we propose a noncontact solution to implement hand gesture recognitions for smart devices. It is based on a continuous wave, time-division-multiplexing (TDM), single-input multiple-output (SIMO) Doppler radar sensor that can be realized by slightly modifying existing RF front ends of smart devices, and a machine-learning algorithm to recognize predefined gestures by classifying deterministic Doppler signals. An experimental setup emulating a smartphone-based radar sensor was implemented, and the experimental results verified the robustness and the accuracy of the proposed approach.

Index Terms—Classification, doppler effect, hand gesture recognition (HGR), machine learning, radar architecture.

I. INTRODUCTION

NOWADAYS, personal devices such as smartphones and tablets have become indispensable for daily human life. Integrated with wireless interfaces, sensors, and powerful computation ability, smart devices are rapidly becoming personal communication, information, and control centers [1].

So far, most interactions between humans and smart devices are based on multitouch screens [2]. With applications of smart devices extending to scenarios such as virtual reality and wireless control of televisions (TVs) and projectors, noncontact interactions based on human hand gesture recognitions (HGRs) have attracted more and more interests [3]–[6].

Originally, noncontact HGR was considered as a human–machine interface (HMI) in computer science [7]. Based on cameras and digital imaging processing algorithms, such a wireless HMI can be classified as an optical approach. Highly relying on computational resources, direct line of sights, and illuminating conditions, the application of computer vision-based HGRs is so far limited [8]. To overcome the above cost, obstacle, and illumination issues, HGRs based on microwave and millimeter-wave approaches, i.e., pulsed and continuous-wave (CW) radar approaches, have been investigated [9]–[18].

Manuscript received June 6, 2020; revised July 21, 2020 and August 12, 2020; accepted September 2, 2020. Date of publication November 4, 2020; date of current version January 5, 2021. This work was supported by NSFC under Grant 61771421 and Grant 62071417. (Corresponding author: Lixin Ran.)

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Digital Object Identifier 10.1109/TMTT.2020.3031619

So far, most of the reported radar-based HGRs are based on the frequency modulated continuous wave (FMCW) scheme [12]–[15], [19], [20]. Among them, the most renowned is the “Google Soli” [15], which is a 60-GHz FMCW radar module with a bandwidth up to 7 GHz. Based on classifying range-Doppler spectrograms [10], it can recognize fine gestures such as finger rubs, which can be used to control features such as the volume and music playlist of a smartphone. However, the added millimeter-wave radar chip and its peripherals increase the price, power consumption, and inner space occupation, all are scarce resources for space and resource constraint devices.

So far, two classes of recognition algorithms have been proposed in the above radar-based solutions. One is based on linear methods, trying to retrieve the displacement of targets, i.e., the moving pattern of the gestures, from the backscattered fields [4]–[6], [9]. The advantage of these methods is that the retrieved gesture patterns are meaningful, having direct correspondence with the actual gestures. The disadvantage is that due to the Mie scattering [21] of hands, the shape of the detected patterns can have significant variation due to different angles of illumination, increasing the difficulty of recognition.

Another class of algorithms is based on artificial intelligence (AI), taking advantage of the nonlinear generalization capacity of artificial neural networks (ANNs) and machine learning-based classifications [22]–[24]. These methods recognize the “signature” of the fields scattered by gestures. The advantage of these methods is that a “black-box” recognition can always be implemented to any predefined gestures, as long as such signatures are used to train the ANN-based classifiers [10], [12]–[14], [16], [17], [19], [25]. For such black-box classifications, sufficient gestures made from different people are required to train the classifiers. Unfortunately, the number of such training gestures can be up to tens of thousands, leading to high human costs and time consumption in gesture data set acquisition, data labeling, and data cleaning.

In this work, we combine the advantages of the above architectures and algorithms, proposing a systematic solution to implement HGRs for smart devices. It is based on a CW, time-division-multiplexing (TDM), single-input multiple-output (SIMO) Doppler radar sensor (DRS) and a machine-learning algorithm to recognize hand gestures by classifying deterministic Doppler signals. Such a DRS can be implemented by slightly modifying the existing RF front ends of smart devices, increasing negligible hardware cost, power consumption, and space occupation. Different from the

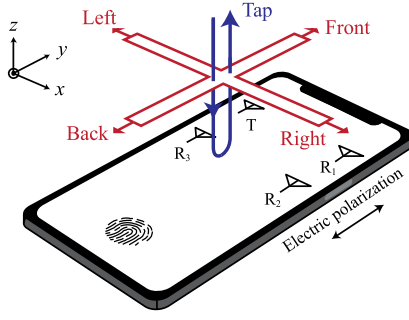
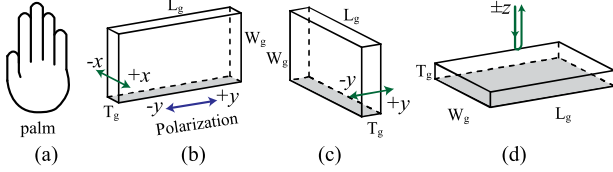


Fig. 2. Definition of the hand gestures.

Fig. 3. Regulation of the hand gestures in y-polarized electric field. (a) Palm. (b) $\pm x$ gesture. (c) $\pm y$ gesture. (d) $\pm z$ gesture.

III. THEORY

A. Definition of Gestures

Fig. 2 shows a smartphone lying on the xy plane. Assuming antennas T, R_1 , R_2 , and R_3 shown in Fig. 1 are placed on the sides of the smartphone. Normally, such antennas are inverted-F antennas polarized in the y -direction.

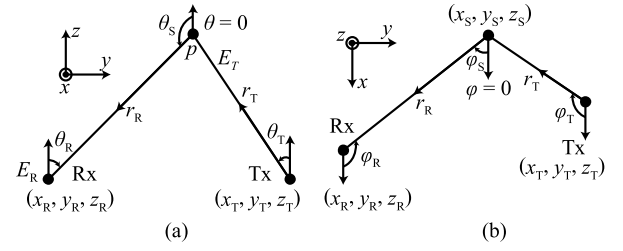
In this work, the hand gestures were defined to realize the major functions of a multitouch screen, i.e., switching between home and subscreens, scrolling pages up and down, and tap actions. For the x - y - z coordinate defined in Fig. 2, the above gestures correspond to the hand movements along the $\pm x$ -, $\pm y$ -, and $\pm z$ -directions over the screen of the smartphone, respectively. They are named “left” (L), “right” (R), “front” (F), “back” (B), and “tap” (TAP) gestures, respectively.

B. Error-Correction Mechanism

In practice, the horizontal gestures can easily deviate from the x - and y -axes, resulting in potential erroneous decisions. To solve this problem, two redundant gestures, i.e., hand movements along the diagonal lines, i.e., $y = x$ and $y = -x$, are defined to maximize the error tolerance. In practice, once a hand gesture is recognized as a diagonal movement, this gesture will be judged as an invalid gesture. Such redundant gestures eliminate the ambiguity regions between the L, R, F, and B gestures, denoted by LF, LB, RF, and RB, respectively.

C. Scattering of a Moving Palm in the Fresnel Near Field

In this work, a 5.8-GHz DRS is used to demonstrate the recognition of such gestures. At this frequency, the corresponding wavelength is 5.2 cm, the same scale as a human hand. In this case, a human palm should be considered as a Mie scatterer, and the analysis should be limited inside the Fresnel near-field range [40].

Fig. 4. Bistatic model for the scattering analysis of a moving palm. (a) yOz view. (b) xOy view.

To simplify the analysis of the Mie scattering of a human hand, we regulate that the gestures will be carried out using a flat palm, as shown in Fig. 3(a). Then, this palm can be approximated as a conductive flat cuboid, moving in the $\pm x$ -, $\pm y$ -, and $\pm z$ -directions with different orientations. In Fig. 3(b)–(d), the length, width, and thickness of the cuboid are denoted by L_g , W_g , and T_g , respectively. The shadowed sides are the ones facing the transmitting and receiving antennas. When the palm is moving in the horizontal and vertical directions, the bottom lateral side and the front side of the cuboid will scatter most of the incident field back to the receiving antennas.

In antenna theory, the scattering of a continuous surface can be considered as the secondary radiation from an aperture antenna. Conventionally, far-field equations such as the Friis equation and the radiation patterns of linear antennas can be conveniently used to analyze such scatterings. However, they are not applicable in the Fresnel near field.

Here, an *ab initio* approach is used to analyze such near-field scattering by integrating the radiation of y -polarized Hertzian dipoles on the rectangular aperture along both the x - and y -directions. In the global x - y - z coordinate shown in Fig. 4, we assume that the phase centers of the transmitting and receiving antenna are located at (x_T, y_T, z_T) and (x_R, y_R, z_R) , respectively. The vertical axes point to the $+z$ -direction. We also assume that the geometric center of the scattering aperture is located at (x_S, y_S, z_S) , with its vertical axis pointing to the $+z$ -direction.

Mathematically, the electric field illuminating the scattering aperture by the transmitting antenna can be expressed as

$$E_T(r_T, \theta_T, \varphi_T) = \hat{y} E_0 G_T(\theta_T, \varphi_T) e^{-jk r_T} / r_T \quad (4)$$

where $G_T(\theta_T, \varphi_T)$ is the radiation pattern of the transmitting antenna and r_T is the distance from transmitting antenna to the element. Similarly, we define $G_R(\theta_R, \varphi_R)$ as the radiation pattern of the receiving antenna, and r_R as the distance from the element to receiving antenna. Here, the far-field radiation patterns can be used because the far-field Fresnel range of quarter-wavelength inverted-F antennas used in smart devices is small.

According to the coordinate systems defined in Fig. 4, for a point on the scattering aperture deviated from (x_S, y_S, z_S) , denoted by $(x_S + \zeta, y_S + \xi, z_S)$, the elevation and azimuth angles and the distances in all the coordinate systems can be

determined as

$$\begin{cases} \theta_T = \arctan\left(\sqrt{(x_S + \zeta - x_T)^2 + (y_S + \zeta - y_T)^2 / (z_S - z_T)}\right) \\ \varphi_T = \arctan[(y_S + \zeta - y_T) / (x_S + \zeta - x_T)] \\ \theta_S = \arctan\left(\sqrt{(x_R - x_S - \zeta)^2 + (y_R - y_S - \zeta)^2 / (z_R - z_S)}\right) \\ \varphi_S = \arctan[(y_R - y_S - \zeta) / (x_R - x_S - \zeta)] \\ \theta_R = \arctan\left(\sqrt{(x_S + \zeta - x_R)^2 + (y_S + \zeta - y_R)^2 / (z_S - z_R)}\right) \\ \varphi_R = \arctan[(y_S + \zeta - y_R) / (x_S + \zeta - x_R)] \\ r_T = \sqrt{(x_S + \zeta - x_T)^2 + (y_S + \zeta - y_T)^2 + (z_S - z_T)^2} \\ r_R = \sqrt{(x_S + \zeta - x_R)^2 + (y_S + \zeta - y_R)^2 + (z_S - z_R)^2} \end{cases} \quad (5)$$

Illuminated by the y -polarized incidence, the current induced on the rectangular scattering aperture satisfies a sinusoidal distribution in the y -direction [40]. It comprises infinite line currents along the x -direction, each with a surface current distribution of

$$J(\zeta, \xi) = I_0 \sin\left[k\left(\frac{l}{2} - |\xi|\right)\right] \quad (6)$$

where the amplitude $I_0 = \sigma_e E_T(r_T, \theta_T, \varphi_T)|_{\zeta=0, \xi=0}$, σ_e being an equivalent conductivity. This current can be further divided into infinite y -polarized Hertzian dipoles. For the Hertzian dipole located at $(x_S + \zeta, y_S + \zeta, z_S)$, its y -component far-field radiation can be derived as (given in the Appendix)

$$\begin{aligned} dE_S^y(\zeta, \xi, r, \theta, \varphi) \\ = \frac{-jk\eta_0 J(\zeta, \xi) e^{-jkr}}{4\pi r} (\cos^2 \theta \sin^2 \varphi + \cos^2 \varphi) d\zeta d\xi. \end{aligned} \quad (7)$$

Then, the total field scattered by the rectangular aperture can be obtained by integrating all the radiations of such Hertzian dipoles along both the x - and y -directions, and the electric field detected by the receiving antenna can be derived as

$$E_R = \int_{-l/2}^{l/2} \int_{-w/2}^{w/2} G_R(\theta_R, \varphi_R) \cdot dE_S^y(\zeta, \xi, r_R, \theta_R, \varphi_R). \quad (8)$$

Fig. 5(a)–(c) shows the θ -dependent amplitude and phase radiation patterns calculated based on (6) at a near-field Fresnel range of 20 cm for the “left/right,” “back/front,” and “tap” gestures, respectively, with L_g , W_g , and T_g being 15, 8, and 2 cm, respectively.

It is seen that the main lobes in all the amplitude patterns are wide, benefiting the coverage of all the receiving antennas. In Fig. 5(a), the phase shift along the $\pm x$ (or $\varphi = 0/180^\circ$, $\theta = 90^\circ$) directions due to the y -polarized illumination is nearly constant and immune to “left/right” motions. Similar flat phase shift regions can also be found in Fig. 5(b) and (c) for the $\pm y$ (or $\varphi = 90^\circ/-90^\circ$, $\theta = 90^\circ$) and $\pm z$ (or $\theta = 0/180^\circ$) directions, which are suitable for the “back/front” and “tap” gestures, respectively. As shown in the following, such flat phase responses will result in quasi-linear, deterministic Doppler phases, which is very important for HGRs. This is also the basis on which we defined the above gestures.

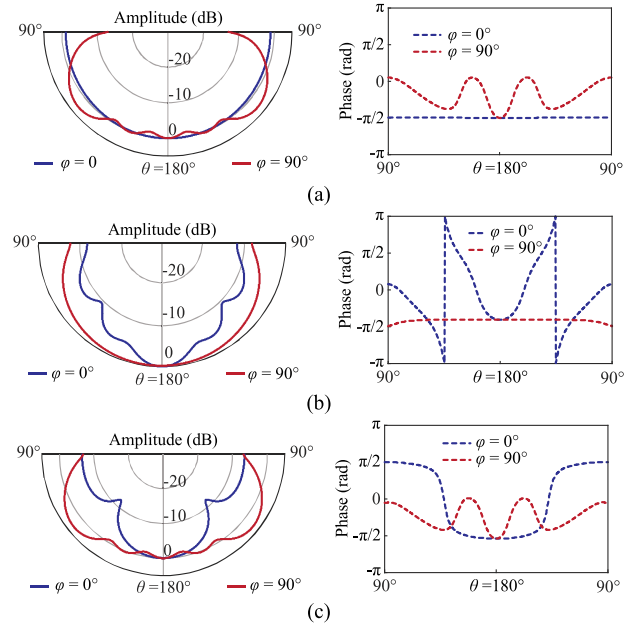


Fig. 5. Scattering patterns calculated by (6) with different aperture size. (a) $w = 2$ cm, $l = 15$ cm. (b) $w = 15$ cm, $l = 2$ cm. (c) $w = 8$ cm, $l = 15$ cm.

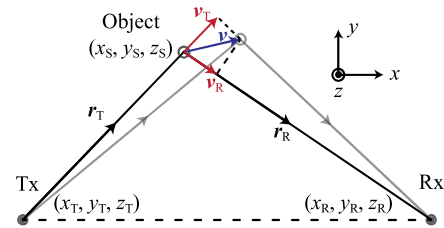


Fig. 6. Doppler phases detected by a bistatic DRS.

D. Doppler Signals Detected by a Bistatic Radar

A SIMO DRS can be treated as a combination of several bistatic radars that share the same transmitting channel. Fig. 6 illustrates a bistatic radar used to detect a moving object. An object moves from (x_S, y_S, z_S) with a velocity $\mathbf{v} = (v_x, v_y, v_z)$ and a displacement $\mathbf{p} = (p_x, p_y, p_z)$. The radial velocities v_T and v_R are the projections of $\mathbf{v}$ onto vectors $\mathbf{r}_T$ and $\mathbf{r}_R$, and thus the roundtrip distance $r(t) = |\mathbf{r}_T(t)| + |\mathbf{r}_R(t)|$.

In a short interval Δt where $\Delta \mathbf{p}(t) = (v_x(t), v_y(t), v_z(t)) \Delta t$

$$r(t + \Delta t) = \sqrt{|\mathbf{r}_T(t) + \Delta \mathbf{p}(t)|^2} + \sqrt{|\mathbf{r}_R(t) - \Delta \mathbf{p}(t)|^2}. \quad (9)$$

Then, the derivative of $r(t)$ can be calculated as (given in the Appendix)

$$\begin{aligned} \frac{dr(t)}{dt} &= \lim_{\Delta t \rightarrow 0} \frac{r(t + \Delta t) - r(t)}{\Delta t} = v(t) \cdot \hat{\mathbf{r}}_T - v(t) \cdot \hat{\mathbf{r}}_R \\ &= |v_T(t)| - |v_R(t)|. \end{aligned} \quad (10)$$

According to the Doppler formula [41], the Doppler frequency shift can be calculated as

$$f_d(t) = (|v_R(t)| - |v_T(t)|) f_c / c \quad (11)$$

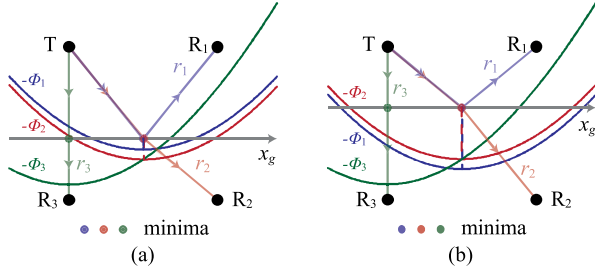


Fig. 7. Variations of Doppler signals detected by the TDM-SIMO DRS in “right” gestures. In the calculations, $a = 2.5$ cm, $h = 10$ cm, x_g represents the gesture moving from -4.5 to 4.5 cm, y_1 and y_2 are the y coordinates of “right” gestures where (a) $y_1 = -0.5$ cm and (b) $y_2 = 0.5$ cm in the examples.

where f_c is carrier frequency and c is the speed of light. Based on (10) and (11)

$$r(t) = -\frac{c}{2\pi f_c} \int 2\pi f_d(t) dt = -\frac{c}{2\pi f_c} \Phi(t). \quad (12)$$

Equation (12) shows the equivalency between the roundtrip distance $r(t)$ and the Doppler signal $-\Phi(t)$.

E. Deterministic Doppler Signals Carried by $\Phi(t)$

For antennas T, R_1 , R_2 , and R_3 arranged in Fig. 2, we assume that their coordinates are $(-a, a, 0)$, $(a, a, 0)$, $(a, -a, 0)$, and $(-a, -a, 0)$, respectively, and a “right” gesture moves on the $z = h$ plane. Its trajectory can be described by $X(x_g, y, h)$, where x_g is the x -coordinate of the scattering phase center of this gesture and y being the constant y -coordinate of the gesture. The roundtrip distances between the transmitter, moving gesture, and the TDM receivers can be expressed as

$$\begin{cases} r_1 = \sqrt{(x_g + a)^2 + (y - a)^2 + h^2} \\ \quad + \sqrt{(x_g - a)^2 + (y - a)^2 + h^2} \\ r_2 = \sqrt{(x_g + a)^2 + (y - a)^2 + h^2} \\ \quad + \sqrt{(x_g - a)^2 + (y + a)^2 + h^2} \\ r_3 = \sqrt{(x_g + a)^2 + (y - a)^2 + h^2} \\ \quad + \sqrt{(x_g + a)^2 + (y + a)^2 + h^2}. \end{cases} \quad (13)$$

Based on (12) and (13), the corresponding Doppler signals $-\Phi_i(t)$ can be calculated, as shown in Fig. 7, where $i = 1, 2, 3$.

It is seen that for the “right” gestures, the minimum of $-\Phi_3(t)$ appears first. Then, those of $-\Phi_1(t)$ and $-\Phi_2(t)$ follow. For antennas located on the corners of a square, the minima of $-\Phi_1(t)$ and $-\Phi_2(t)$ will appear simultaneously. Such deterministic but different Doppler signals also appear for other gestures. In this work, the temporal order of such minima of $-\Phi_i(t)$ will be used in robust gesture recognition.

In (13), h also represents the height of the gestures. When h is changed, the shape of the $-\Phi_i(t)$ curves will also be changed. However, the temporal order of the minima will remain unchanged. It means that gesture recognitions based on the temporal order of such minima will be independent on the height of gestures.

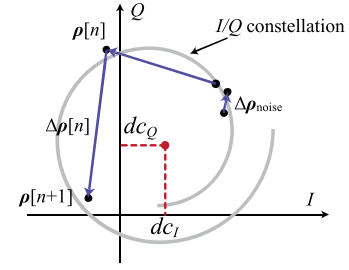


Fig. 8. Illustration of $\Delta \rho[n]$ and $\Delta \rho_{\text{noise}}$ in the I/Q constellation.

F. Segmentation of Detected Signals

To recognize the predefined gestures, the signals containing all the motion information, i.e., $\Phi[n]$, should be isolated first. This can be achieved by a dynamic power analysis to the detected I/Q signals.

Fig. 8 illustrates the vector I/Q signals at sampling n and $n + 1$. The variation of the I/Q signals during the adjacent sampling is denoted by the vector $\Delta \rho[n]$. In the presence of a moving scatterer,

$$\begin{aligned} \Delta \rho[n] &= \begin{pmatrix} A_I[n+1] \cos(\Phi[n+1]) - A_I[n] \cos(\Phi[n]) + \Delta n_I[n] \\ A_Q[n+1] \sin(\Phi[n+1]) - A_Q[n] \sin(\Phi[n]) + \Delta n_Q[n] \end{pmatrix}^T \\ &= \begin{pmatrix} A_I[n+1] \cos(\Phi[n+1]) - A_I[n] \cos(\Phi[n]) + \Delta n_I[n] \\ A_Q[n+1] \sin(\Phi[n+1]) - A_Q[n] \sin(\Phi[n]) + \Delta n_Q[n] \end{pmatrix}^T \end{aligned} \quad (14)$$

reflecting the “differential power” $\Delta P[n]$ of the I/Q signals, where $\Delta n_I[n]$ is $n_I[n+1] - n_I[n]$, so it is with $\Delta n_Q[n]$. Here, $\Delta P[n]$ can be calculated as

$$\Delta P[n] = |\Delta \rho[n]|^2. \quad (15)$$

In the absence of any motion, $\Delta \rho_{\text{noise}}[n] = (\Delta n_I[n], \Delta n_Q[n])$, representing the “differential power” of the circuit noise. Clearly, $\Delta \rho$ is always larger than $\Delta \rho_{\text{noise}}$, and thus, ΔP can be effectively used to separate the segments containing the gesture information from the detected signals.

To determine the threshold of the separation, the mathematical expectation of $\Delta \rho_{\text{noise}}$, i.e., $E(\Delta \rho_{\text{noise}})$, can be determined, based on which a value slightly larger than this expectation can be chosen as the threshold. In this work, this value is $1.1 E(\Delta \rho_{\text{noise}})$.

To ensure the reliability of the separation, an m -point moving average window can be performed to $\Delta P[n]$ at each time interval with

$$\overline{\Delta P_m}[n] \triangleq m^{-1} \sum_{j=n-m/2+1}^{n+m/2} \Delta P[j] \quad (16)$$

to reduce potential erroneous judgments due to unexpected glitches. If $\overline{\Delta P_m}[n]$ is larger than the threshold, a motion will be considered to exist inside this time interval. All such continuous time intervals correspond to a signal segment, during which a motion is considered to exist. The same operation can be performed to signals detected by all the receiving channels. The obtained three signal segments reflect the same motion detected from different orientations. The union of such segments will result in a widest segment that

containing the information of a complete motion, from which a potential gesture can be recognized.

G. Data Alignment for TDM Samplings

The utility of the TDM-SIMO architecture shown in Fig. 1 significantly reduces the complexity and cost of the DRS. However, the sampled data sequences need to be aligned.

For M TDM receiving channels, the detected serial I -channel sequences $I_1[n]$, $I_2[n+1/M]$, ..., $I_M[n+(M-1)/M]$, $I_1[n+1]$, $I_2[n+1+1/M]$, ... should be realigned into $I_1[n]$, $I_2[n]$, ..., $I_M[n]$, $I_1[n+1]$, $I_2[n+1]$, ... The difference between the detected and aligned signals for the i th channel is $(i-1)T_{\text{TDM}}/M$, where T_{TDM} is the interval between adjacent TDM sampling. In this work, a set of digital phase shift filters [42] are designed to compensate this temporal asynchrony, so that the z -domain response of this filter for i th channel can be obtained as is $H_i(z) = z^{(i-1)/M}$. Its frequency response is

$$H_i(e^{j\omega}) = H_i(z)|_{z=e^{j\omega}} = e^{j(i-1)\omega/M} \quad (17)$$

and such all-pass phase shift filter can be synthesized as

$$\begin{aligned} h_i(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{(i-1)j\omega/M} e^{j\omega k} d\omega \\ &= \frac{\sin\{\pi[n + (i-1)/M]\}}{\pi[n + (i-1)/M]}. \end{aligned} \quad (18)$$

After convoluting $I_i[n]$ with $h_i(n)$ correspondingly, the data sampled from different I -channels can be aligned. Similarly, the detected Q -channel signals can also be aligned.

H. Rescaling and Fitting of Segmented Gestures

After the segmentation and data alignment, the consecutive temporal segments, each containing a single gesture, can be obtained. For the widest gesture segment, its actual duration, or the total number of samples, varies from gesture to gesture. In addition, the detected original Doppler phase signal Φ_i can be zigzagged and, therefore, rescaling and fitting to the segmented gesture signals are needed, to obtain regulated gesture signals with the same length and smoothed shapes for further recognition.

In this work, all the widest segments are rescaled into (0, 1). To maintain the shape of $-\Phi_i$ and prevent the impact from glitches, in this work, an eighth-order polynomial fitting is adopted so that the noise on signal $-\Phi_i$ would not be fit.

After the rescaling and fitting, the moments of the minima of $-\Phi_i$ for the i th channel discussed in Section II-E can be quantified between the interval (0, 1) and denoted by $t_{m,i}$. In this work, the moments for the minima are denoted in vector form as $\mathbf{t}_m = (t_{m,1}, t_{m,2}, t_{m,3})$, which can characterize each gesture and will be the input of the classifier discussed in the following.

After this rescaling, the minima moments become independent on the speed of gestures. It plays an important role in recognizing both fast and slow gestures.

I. Classification-Based Gesture Recognition

Classification is one of the commonly used techniques in machine learning [23], aiming to categorize a set of data

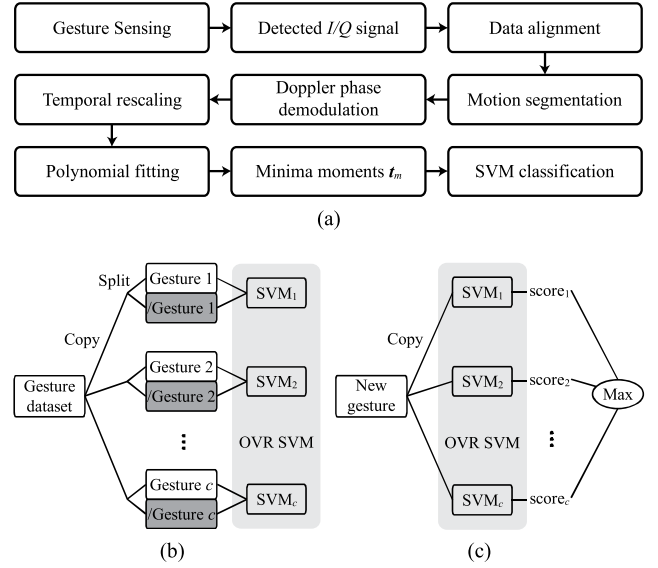


Fig. 9. Flowchart of the gesture recognition. (a) Signal processing. (b) Training of the OVR SVM. (c) Gesture recognition based on trained OVR SVM.

into classes. It has been used in image recognition, speech processing, bioinformatics, natural language processing, and so on [24].

So far, a variety of classification algorithms, or classifiers, has been proposed. Mainstream classifiers include logistic regression, linear discriminant analysis (LDA), decision trees, naïve Bayes classifier, and support vector machine (SVM) [43].

In this work, the one versus rest (OVR) SVM designed for multiclassification is chosen to recognize the predefined multiple gestures. The flowcharts of the signal processing, the training of the OVR SVM based on known gesture data set, and the recognition for unknown gestures are shown in Fig. 9(a)–(c), respectively.

To train the OVR SVM, the data set comprises known gestures can be prepared by detecting each defined gesture for N times. Then, this data set can be duplicated into c copies. Here, c is the number of the defined gestures. In the training of the i th SVM, all the i th gestures are labeled as +1, and all the rest are labeled as -1.

Based on the trained SVMs, a new gesture can be recognized by comparing the “score,” or similarity, evaluated by each SVM. If the highest score was evaluated by the i th SVM, this gesture will be recognized as the i th one. Details can be found in [23].

IV. SIMULATIONS

To verify the effectiveness of the methods proposed in Section III, we ran numerical simulations based on derived equations following the steps shown in Fig. 9. In all the simulations, the coordinates of the transmitting and receiving antennas T , R_1 , R_2 , and R_3 are $(-2.5, 2.5, 0)$, $(2.5, 2.5, 0)$, $(2.5, -2.5, 0)$, and $(-2.5, -2.5, 0)$; L , W , and T of the simplified palm are 15, 8, and 2, all in cm, respectively.

A. Doppler Signals Detected of Different Palm Orientation

The defined gestures can be classified into two types. For the “left,” “right,” “front,” “back,” and “diagonal” gestures,

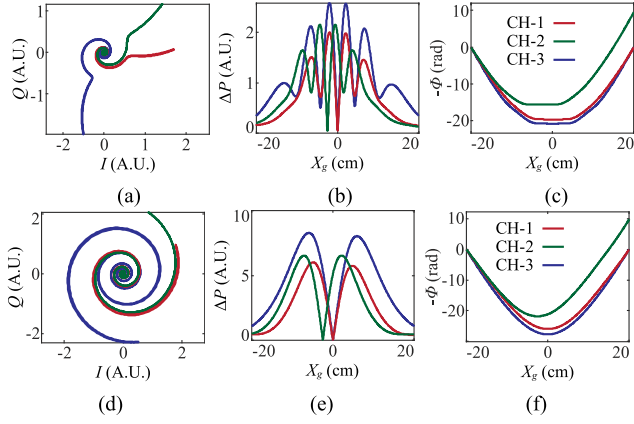


Fig. 10. Doppler signals simulated for different orientations of the palm. (a)–(f) are the constellation of the I/Q signals, the differential power for the segmentations, and the retrieved Doppler phases for the parallel and vertical palms, respectively.

the palm is vertical to the antenna plane, scattering the illuminating field with a narrow aperture. For the “tap” gesture, the palm is parallel to the antenna plane, scattering with a much wider aperture.

To explain why these two types of gestures were defined, we compared the Doppler signals calculated for “right” gestures performed with both vertical and parallel palms. In the calculation, the centers of both scattering apertures, i.e., (X_g, y, h) , move from $(-20, 0, 10)$ to $(20, 0, 10)$, respectively. Based on (2)–(8), the constellation of the I/Q signals, the differential power, $\Delta P[n]$, for the segmentations, and the retrieved Doppler phases, $-\Phi[n]$, are plotted in Fig. 10.

It can be seen that the moving vertical and parallel palms can result in very different I/Q signals and Doppler phases. Especially, for the parallel moving palm, the minima in Fig. 10(c) can hardly be determined. However, the minima in Fig. 10(f) can be easily identified. This is the consideration of the definition of gestures moving parallel above the antenna plane.

B. Simulations Following the Flowchart

In this section, numerical simulations will be performed following the steps shown in Fig. 9(a).

First, using the same method applied in calculating Fig. 10, the I/Q signals resulting from two successive “right” and “left” gestures “detected” by the R_1 , R_2 , and R_3 antenna can also be calculated. The results are shown in Fig. 11. The difference is that a white Gaussian noise is added so that the I/Q signals can be generated with a 20-dB signal-to-noise ratio (SNR).

Then, such signals are segmented according to Section III-F. Panels I and II in Fig. 12 show the demodulated Doppler phases and the 60-point moving averaged differential power $\overline{\Delta P_{60}}[n]$, respectively. Choosing the threshold as 2, the gestures can be segmented in regions marked in gray. It is seen that both gestures are properly segmented. Panel III shows the same calculation as panel II under a worse 10-dB SNR condition. It is seen that the gesture segments can be hardly identified.

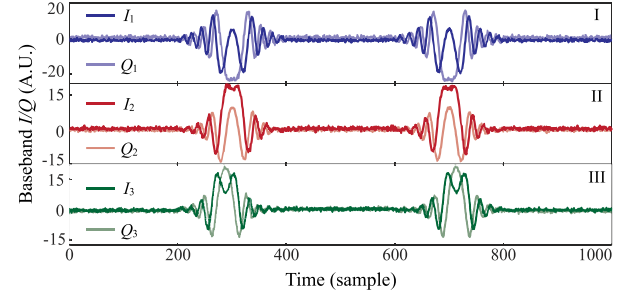


Fig. 11. Simulated I/Q signals result from two successive “right” and “left” gestures “detected” by the R_1 , R_2 , and R_3 antennas, respectively.

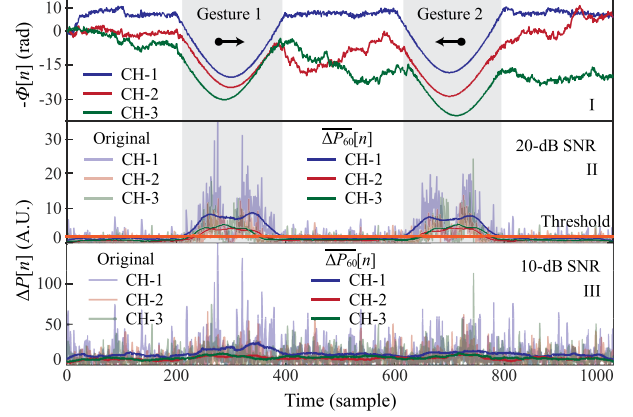


Fig. 12. Segmentations of two successive “right” and “left” gestures. Panel I, demodulated Doppler phase under 20-dB SNR. Panel II, $\Delta P[n]$ and $\overline{\Delta P_{60}}[n]$ of simulated baseband signals under 20-dB SNR, as well as a threshold for segmentation. Panel III, $\Delta P[n]$ and $\overline{\Delta P_{60}}[n]$ of simulated baseband signals under 10-dB SNR.

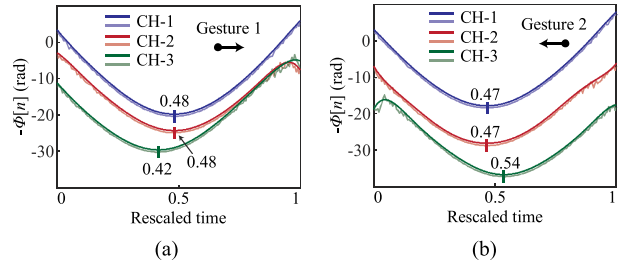


Fig. 13. Determination of Doppler minima in the rescaled and fit Doppler phases for the successive “right” and “left” gestures. Dark lines: eighth-order polynomial fit Doppler phase. Light lines: original Doppler phase with noises. The quantified minima moments for (a) gesture 1 are $t_m = (0.48, 0.48, 0.42)$, and the that for (b) gesture 2 is $t_m = (0.47, 0.47, 0.54)$.

Next, the segmented gestures are rescaled based on the discussion in Section III-H with an eighth-order polynomial fitting. After the rescaling and fitting, as shown in Fig. 13, the minima’s moments of the rescaled Doppler phases can be clearly extracted, based on which the temporal sequential order of these minima can also be determined. In this work, such deterministic, quantified minima in the order of channels, e.g., $t_m = (0.48, 0.48, 0.42)$ and $t_m = (0.47, 0.47, 0.54)$ in this figure, are the inputs of the OVR SVM classifier, instead of the I/Q constellations or the Doppler signatures of used in the “black-box” gesture classifications.

Fig. 14 shows the Doppler phases segmented from I/Q signals simulated for all the defined gestures. In the simulation,

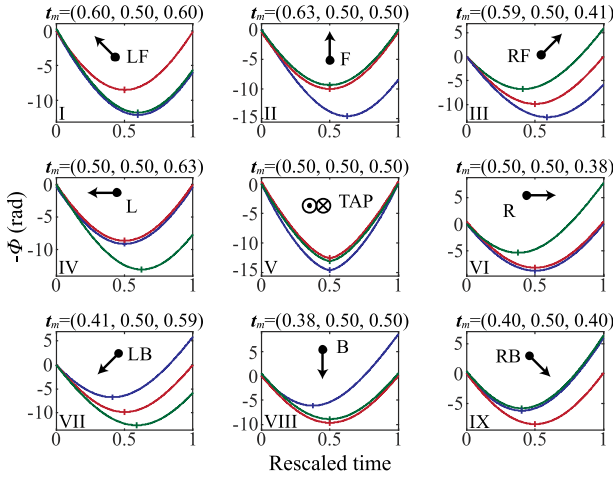


Fig. 14. Simulated Doppler phases of defined gestures detected by different receiving channels.

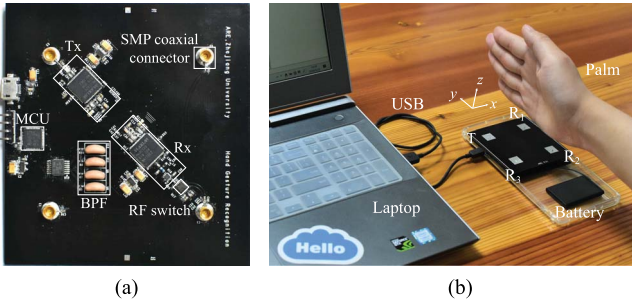


Fig. 15. TDM-SIMO DRS implemented with a pair of Wi-Fi chips. (a) RF board. (b) Battery powered DRS and the experimental environment.

for all the gestures moving in horizontal directions, the distance between the bottom side of the hand and the antenna plane is set as 10 cm, and for the tap gesture, the shortest distance between the palm and the antenna plane is set as 5 cm. It is seen that the temporal sequential orders of minima for each segmented gesture is deterministic.

V. EXPERIMENTS

A. Implementation of the TDM-SIMO DRS

Fig. 15 shows the TDM-SIMO DRS implemented according to the architecture proposed in Fig. 1. In this work, a pair of Wi-Fi chips, Maxim Integrated MAX2829s, were used to emulate the 5.8-GHz LTE-U and Wi-Fi channels, and an RF switch, HMC7992, was used to switch the receiving channel to antennas R_1 , R_2 , and R_3 , respectively. Similar to [44], the L-shaped arrangement of receiving antennas was used to detect 3-D gestures. Without losing any generality, such antennas were designed as standard patch antennas. The antenna and the RF boards were connected by four sub-miniature push-on connector (SMP) coaxial connectors. A standard lithium battery was used to supply power to the DRS. An STM32F103T8U6 micro-controller unit (MCU) was used for Wi-Fi chips configuration and I/Q signals acquisition by the MCU with its built-in ADCs. The sampled data were finally transferred to a laptop via an universal serial bus (USB)-universal asynchronous receiver/transmitter (UART) bridge chip, Silicon Labs CP2102, mounted on the RF board.

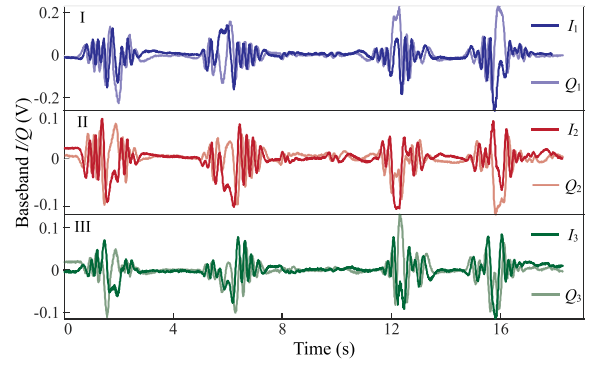


Fig. 16. Experimental I/Q signals result from four successive “front,” “back,” “left,” and “right” gestures detected by antennas R_1 , R_2 , and R_3 , respectively.

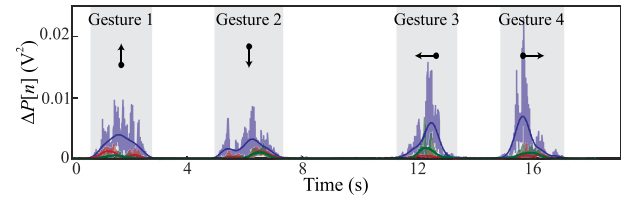


Fig. 17. Segmentation process on the signals detected in Fig. 16 based on the differential power $\Delta P[n]$ as well as moving averaged $\Delta P_{60}[n]$.

B. Processing of Experimental Gesture Signals

Corresponding to Figs. 11–13, the panels I–III in Fig. 16 show the baseband I/Q signals detected by antennas R_1 , R_2 , and R_3 , respectively, for successive “front,” “back,” “left,” and “right” gestures, respectively. Using the same approach for calculating Figs. 12 and 17 shows the segmented gestures detected in Fig. 16. Fig. 18 shows the demodulated Doppler phases and their minima moments t_m for further classification.

It is seen that for experimental data, taking the “right” gesture as the example, the minima of $-\Phi_1(t)$ and $-\Phi_2(t)$ can appear at slightly different times due to the nonplanar scattering aperture of the actual hand. However, if the data set used to train the classifier included such deviated minima, it has few impacts on the correction of the classification.

Corresponding to Fig. 14, the same nine gestures were performed with the same distances to the antenna plane, and the constellations of I/Q signals detected for such gestures by the TDM-SIMO DRS are shown in Fig. 19, respectively. It can be noticed that both the amplitude and the shape of the “tap” gesture are significantly different from the others due to the vertical movements and the larger scattering area of the palm.

Fig. 20 shows the $-\Phi_i$ retrieved from the constellations plotted in Fig. 19. It is seen that the sequential orders of the minima moments t_m are consistent with those in Fig. 14.

C. Gesture Classifications

To classify the defined gestures, we collected 500 samples for different gestures as the data set, 50–60 samples for each gesture. Among them, 80% in the 500 samples (around 40–50 samples for each gesture) were randomly

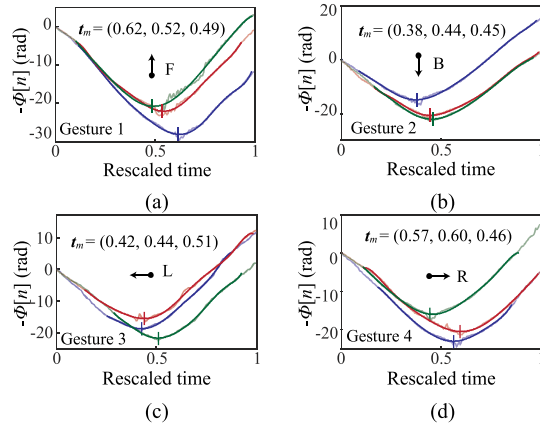


Fig. 18. Determination of Doppler minima of the rescaled and fit Doppler phases for the four successive gestures in Fig. 16. Dark lines: eighth-order polynomial fit Doppler phase curves. Light lines: original Doppler phase curves. The quantified minima moments for the four gestures are (0.62, 0.52, 0.49), (0.38, 0.44, 0.45), (0.42, 0.44, 0.51), and (0.57, 0.60, 0.46), respectively.

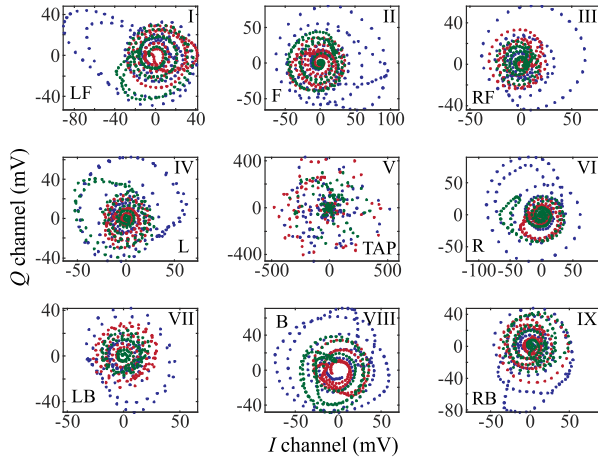


Fig. 19. Experimental constellations of performed gestures.

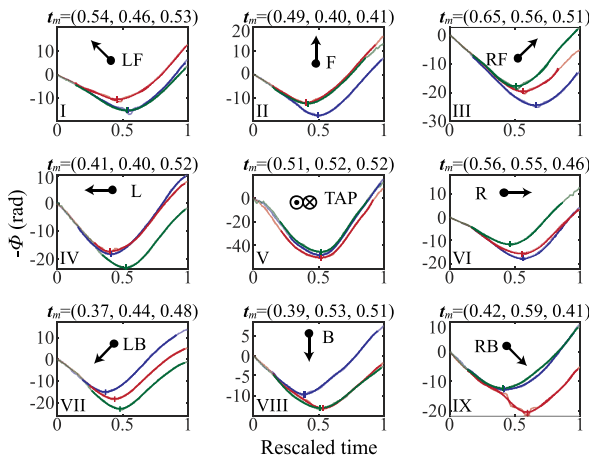


Fig. 20. Experimental Doppler phase variations and minima moments of performed gestures.

chosen as the training set. The rest 20% were chosen as the testing set, including unknown gestures to be recognized. To visualize the minima moments t_m in the training and testing sets, a rectangular coordinate is established, whose axes

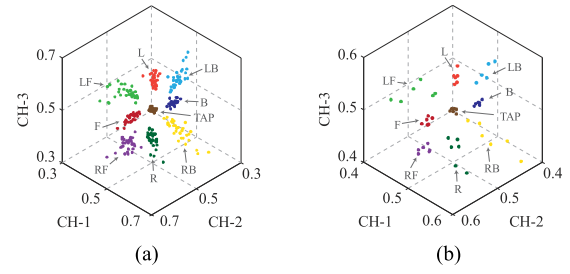


Fig. 21. Mapping of Doppler phase minima moments t_m of all kind of gesture into the 3-D feature coordinate. The axes represent the rescaled time for channel 1–3. (a) Training set. (b) Testing set.

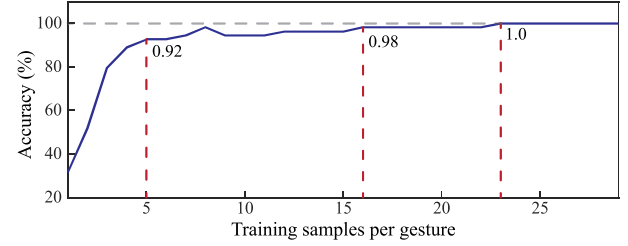


Fig. 22. Learning curve of the OVR SVM evaluated by the unknown samples.

correspond to the minima moments detected by each receiving channel. As shown in Fig. 21, the minima moments of each kind of gesture are clustered and clearly separated from other clusters in both data sets, verifying the determinacy of gestures represented by the minima moments. It also implies that using the OVR SVM trained by the training set, the gestures in the testing set can be accurately classified.

To evaluate the robustness of the minima moments-based classification, we calculate the learning curve [23] by classifying 50 unknown gestures based on the OVR SVM trained by different subsets of the data set. The result is shown in Fig. 22. It is seen that a 98% recognition accuracy can be recognized by the OVR SVM trained by only six samples, and the 100% accuracy can be obtained when the training samples were increased to 23. The training only costed less than 3 s using a regular personal computer.

To verify the generalizability of this classifier, a one-subject-leave-out experiment [12] was conducted by inviting four additional subjects to perform the same gestures. Each gesture is repeated for 10 times. Before performing, they were aware of the moving direction of each gesture, the posture of the hand, and the sensitive region. Fig. 23 shows the mapping of the data sets consisting of the minima moments detected from these subjects. It is seen that compared with the mapping of the training set shown in Fig. 21(a), these minima moments are also clustered and clearly separated. It indicates that although differences exist between the gestures performed by different subjects, the rescaled minima moments exhibit sufficient similarity for the OVR SVM classifier.

Using the classifier trained by the data set detected from the original subject, the minima moments detected from additional subjects shown in Fig. 23 were classified. The recognition accuracies 96.67%, 100%, 97.78%, and 96.67% can be obtained by calculating the confusion matrix for

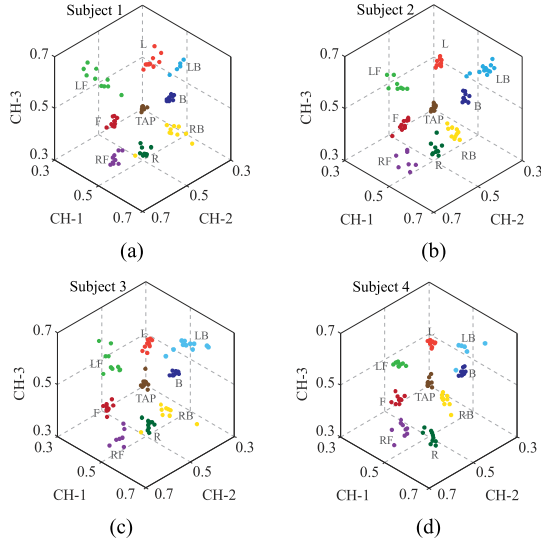


Fig. 23. Mapping of the minima moments detected from the new subjects.

Tr \ Pr	F	B	RF	RB	R	LF	LB	L	TAP
F	24	0	0	0	0	0	0	0	0
B	0	24	0	0	0	0	0	0	0
RF	0	0	22	0	0	0	0	0	2
RB	0	0	0	24	0	0	0	0	0
R	0	0	0	0	24	0	0	0	0
LF	0	0	0	0	0	24	0	0	0
LB	0	0	0	0	0	0	24	0	0
L	0	0	0	0	0	0	0	23	1
TAP	0	0	0	0	0	0	0	0	24

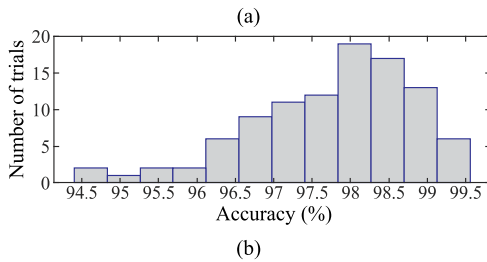


Fig. 24. Accuracy evaluation on the trained OVR SVM. (a) Confusion matrix, in which Tr and Pr stand for “truth” and “prediction,” respectively. (b) Histogram of cross-validation accuracy in 100 trials.

each new subject, respectively. This result verifies that the classifier trained by a certain subject is applicable for unknown subjects.

Finally, the cross-validation [23] experiment was conducted, in which the gestures detected from all the five subjects were mixed. Among the mixed gestures, 75% gestures were randomly chosen as the training set, and the remaining 25% ones were used as testing set. For each gesture, the number of gestures in the testing set is 24. Fig. 24 shows the confusion matrix calculated for gestures performed by all the five subjects attended the one-subject-leave-out experiment. It can be calculated that the total recognition accuracy is up to 98.6%. Also, Fig. 24(b) shows the histogram of cross-validation accuracies in 100 trials. It is calculated that the maximum, minimum, and average accuracies are 99.54%, 94.44%, and 97.83%, respectively, with a standard deviation of 1.15%.

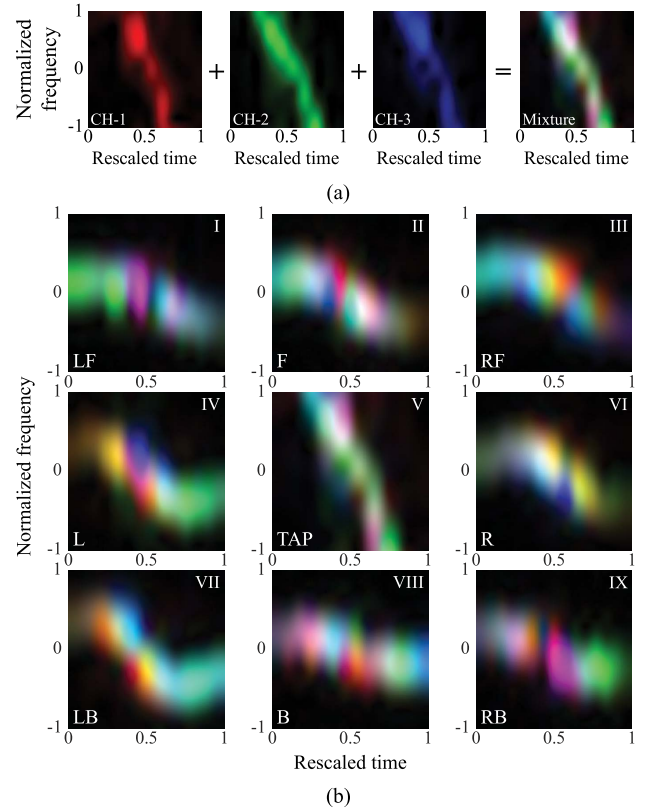


Fig. 25. Time-frequency spectrograms resulted from different gestures. (a) Monochromatic and panchromatic spectrograms. (b) Panchromatic Doppler spectrograms used for DCNN-based classification.

D. Discussion

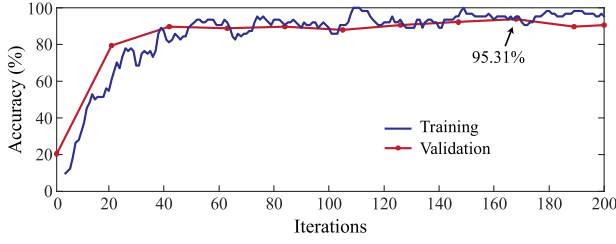
The above results verified the effectiveness and accuracy of the approach proposed for smart devices based on detecting and processing of deterministic, multichannel Doppler signals.

As the consumer products, smart devices, especially smart-phones, are highly integrated, compact, and extremely concern about battery life, product size, and cost. Working at a single frequency, this CW solution can be implemented by reusing the existing RF resources of smart devices, increasing negligible cost, power consumption, and space occupation.

A unique characteristic resulted from the determinacy of the Doppler signals is the high recognition accuracy achieved by the highly efficient OVR SVM classifier trained by a very small data set. In comparison, much more training samples would be needed in the gesture recognition based on classifying the time-frequency spectra of the Doppler signals.

For comparison, Fig. 25(a) shows the short-time Fourier transform (STFT) of the Doppler phases detected by each channel of the SIMO DRS. In the calculation of this STFT spectrogram, a 20-point Hanning sliding window is used for each channel, and the number of overlapping points is 10. Since the temporal duration of each gesture is different, the spectrogram is resized into 400×400 pixels. Then, the spectrogram intensity of each channel is normalized into (0, 1).

The obtained time-frequency spectra were plotted into monochromatic red, green, and blue colors. Then, they are mixed to obtained a panchromatic spectrogram. Fig. 25(b)



(a)

Pr \ Tr	F	B	RF	RB	R	LF	LB	L	TAP
F	24	0	1	0	0	0	0	0	0
B	0	25	0	0	0	0	1	0	0
RF	0	0	21	0	1	0	0	0	0
RB	0	0	2	12	2	1	0	0	0
R	0	0	0	0	22	0	0	0	0
LF	0	0	0	0	0	21	0	0	0
LB	0	0	0	0	0	0	19	0	0
L	0	0	0	0	0	0	1	19	0
TAP	0	0	0	0	0	0	0	0	20

(b)

Fig. 26. Comparison to the DCNN-based classification. (a) Training of the DCNN classifier. (b) Confusion matrix.

shows the spectrograms of the nine different gestures calculated based on the signals shown in Fig. 19. In previous Doppler radar-based recognitions [10], [12], [14], [16], [17], [25], such “images” were used to train the classifiers, normally a deep convolutional neural network (DCNN), and then unknown gestures can be recognized by the trained DCNN.

As an example, 560 spectrograms of detected gestures were used to train the “MobileNets” DCNN intended for mobile and embedded devices [45], and used a mini-batch size of 8 to avoid potential overfitting. Such spectrograms come from the mixture of all the five subjects. Then, 192 unknown spectrograms were classified by the trained DCNN so that the recognition accuracy with respect to the iteration time can be obtained.

As seen in Fig. 26(a), the recognition accuracy increases overall with the increased iteration number. The final recognition accuracy reaches 95.31%. Using a regular computer, the training of the DCNN based on the spectrograms costed around 8 min, much longer than the 3 s costed in the training of the OVR SVM based on the sequenced minima moments. While the DCNN would require a much larger training set, it is difficult to reach a 100% recognition accuracy. Based on the trained DCNN, the confusion matrix can be calculated, as shown in Fig. 26(b).

Another challenge for practical gesture recognitions is to deal with interfering random gestures and motions. In this work, while the OVR SVM can help to filter out some of them, this problem is solved during the signal segmentation. As shown in Fig. 20, for each qualified gesture, since the segmented curve of the Doppler phase detected by each receiving antenna is convex, there is only one minimum moment. Therefore, the random gestures or motions can be excluded by simply examining the number of minima moments in segmented signals.

Finally, the gestures defined in this work correspond to the major functions of multitouch screens. The purpose is to provide additional wireless gesture control functions for smart devices. The Doppler phase curves of such gestures are convex functions. Our method can also be extended to bidirectional and multidirectional gestures, whose Doppler phase curves consist of connected convex functions. Such gestures can be a combination of gestures performed in the horizontal or vertical plane. However, the existing method does not apply for gestures from which sequenced minima moments cannot be retrieved. For these gestures such as finger rubs, micro-Doppler-based analyses may be indispensable.

VI. CONCLUSION

In conclusion, we theoretically analyzed and experimentally demonstrated HGR based on a SIMO-DRS that can be implemented in smart devices by slightly modifying their existing RF front ends. Based on detecting and processing deterministic Doppler signals detected by the SIMO radar system, a small data set can be used to train an OVR SVM and realize millisecond recognition of nine predefined gestures. We envision this robust, highly efficient approach can also be extended to a variety of applications.

APPENDIX

A. Derivation of (7)

For a y-polarized Hertzian dipole, denote vector potential as

$$\mathbf{A} = \hat{\mathbf{y}} \mu I \Delta l e^{-jkr} / 4\pi r \quad (\text{A.1})$$

where Δl and I are an infinitesimal length and a linear current intensity, respectively. In an r - θ - φ spherical coordinate

$$\hat{\mathbf{y}} = \hat{\mathbf{r}} \sin \theta \sin \varphi + \hat{\boldsymbol{\theta}} \cos \theta \sin \varphi + \hat{\boldsymbol{\phi}} \cos \varphi. \quad (\text{A.2})$$

Then, the electric field

$$\begin{aligned} \mathbf{E} &= \frac{1}{j\omega\epsilon} \nabla \times \mathbf{H} = \frac{1}{j\omega\epsilon} \nabla \times \left(\frac{1}{\mu} \nabla \times \mathbf{A} \right) \\ &= \frac{1}{j\omega\epsilon} \frac{I \Delta l}{4\pi} \left[\frac{2}{r^3} (1 + jkr) \sin \theta \sin \varphi e^{-jkr} \hat{\mathbf{r}} \right. \\ &\quad \left. + \left(\frac{k^2}{r} - \frac{1}{r^3} - \frac{jk}{r^2} \right) \cos \theta \sin \varphi e^{-jkr} \hat{\boldsymbol{\theta}} \right. \\ &\quad \left. + \left(\frac{k^2}{r} - \frac{1}{r^3} - \frac{jk}{r^2} \right) \cos \varphi e^{-jkr} \hat{\boldsymbol{\phi}} \right]. \quad (\text{A.3}) \end{aligned}$$

In the far-field range

$$\begin{aligned} \lim_{r \rightarrow \infty} \mathbf{E} &= \frac{1}{j\omega\epsilon} \frac{I \Delta l k^2}{4\pi r} e^{-jkr} (\hat{\boldsymbol{\theta}} \cos \theta \sin \varphi + \hat{\boldsymbol{\phi}} \cos \varphi) \\ &= \frac{-jk\eta_0 I \Delta l e^{-jkr}}{4\pi r} (\hat{\boldsymbol{\theta}} \cos \theta \sin \varphi + \hat{\boldsymbol{\phi}} \cos \varphi). \quad (\text{A.4}) \end{aligned}$$

In cartesian coordinate

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= \hat{\mathbf{x}} \cos \theta \cos \varphi + \hat{\mathbf{y}} \cos \theta \sin \varphi - \hat{\mathbf{z}} \sin \theta \\ \hat{\boldsymbol{\phi}} &= -\hat{\mathbf{x}} \sin \varphi + \hat{\mathbf{y}} \cos \varphi. \quad (\text{A.5}) \end{aligned}$$

Then, the y -component of the electric field scattered by the Hertzian dipole is

$$E_{\text{dipole}}^y = \frac{-jk\eta I \Delta l e^{-jkr}}{4\pi r} (\cos^2 \theta \sin^2 \varphi + \cos^2 \varphi). \quad (\text{A.6})$$

Replacing $I \Delta l$ by the actual surface current element $J(\zeta, \xi) d\zeta d\xi$, the far-field electric field radiated from this Hertzian dipole can be derived as

$$\begin{aligned} dE_S^y(\zeta, \xi, r, \theta, \varphi) \\ = \frac{-jk\eta_0 J(\zeta, \xi) e^{-jkr}}{4\pi r} (\cos^2 \theta \sin^2 \varphi + \cos^2 \varphi) d\zeta d\xi. \end{aligned} \quad (\text{A.7})$$

B. Derivation of (9) and (10)

For the bistatic radar described in Fig. 6, the transient displacement $\mathbf{p} = (p_x, p_y, p_z)$ and the 3-D transient velocity $\mathbf{v} = (v_x, v_y, v_z)$ satisfy $\mathbf{v} = d\mathbf{p}/dt$, whose differential form is $d\mathbf{p}(t) = \mathbf{v}(t)dt$. Therefore, the roundtrip distances at the time t and $t + \Delta t$ can be calculated as $r(t) = |\mathbf{r}_T(t)| + |\mathbf{r}_R(t)|$ and

$$r(t + \Delta t) = \sqrt{|\mathbf{r}_T(t) + \Delta \mathbf{p}(t)|^2} + \sqrt{|\mathbf{r}_R(t) - \Delta \mathbf{p}(t)|^2} \quad (\text{A.8})$$

respectively. For Δt approaching to 0

$$\begin{aligned} \lim_{\Delta t \rightarrow 0} r(t + \Delta t) \\ = \lim_{\Delta t \rightarrow 0} \sqrt{|\mathbf{r}_T(t)|^2 + 2\mathbf{r}_T(t) \cdot \Delta \mathbf{p}(t) + |\Delta \mathbf{p}(t)|^2} \\ + \sqrt{|\mathbf{r}_R(t)|^2 - 2\mathbf{r}_R(t) \cdot \Delta \mathbf{p}(t) + |\Delta \mathbf{p}(t)|^2} \\ \approx \lim_{\Delta t \rightarrow 0} \sqrt{|\mathbf{r}_T(t)|^2 + 2\mathbf{r}_T(t) \cdot \Delta \mathbf{p}(t)} \\ + \sqrt{|\mathbf{r}_R(t)|^2 - 2\mathbf{r}_R(t) \cdot \Delta \mathbf{p}(t)} \\ \approx \lim_{\Delta t \rightarrow 0} |\mathbf{r}_T(t)| + \Delta \mathbf{p}(t) \cdot \hat{\mathbf{r}}_T(t) + |\mathbf{r}_R(t)| - \Delta \mathbf{p}(t) \cdot \hat{\mathbf{r}}_R(t). \end{aligned} \quad (\text{A.9})$$

Note that in the first approximation, $|\Delta \mathbf{p}(t)|^2$ has been omitted. In the second approximation, the second-order Taylor expansion $\lim_{\alpha \rightarrow 0} (1 + \alpha)^{1/2} \approx 1 + \alpha/2$ is applied. Thus, the derivative of the roundtrip distance can be derived as

$$\begin{aligned} \frac{dr(t)}{dt} &= \lim_{\Delta t \rightarrow 0} \frac{r(t + \Delta t) - r(t)}{\Delta t} \\ &= (\Delta \mathbf{p}(t) \cdot \hat{\mathbf{r}}_T - \Delta \mathbf{p}(t) \cdot \hat{\mathbf{r}}_R) / \Delta t \\ &= \mathbf{v}(t) \cdot \hat{\mathbf{r}}_T - \mathbf{v}(t) \cdot \hat{\mathbf{r}}_R \\ &= |v_T(t)| - |v_R(t)|. \end{aligned} \quad (\text{A.10})$$

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